

DTV Guide

July 2000

ON THE
ROAD
to DTV

The resource for information on the analog to digital transition

A joint production by





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by CEA President and CEO Gary Shapiro

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DIGITAL TELEVISION – THE NEXT CHALLENGE

After 75 years of good ole NTSC, it hasn't been easy shifting to a new television standard. Thankfully, the sheer "oh my gosh" beauty of digital television (DTV) will ensure its success even though so many established forces work to delay or derail it.

The biggest threat today to DTV comes from the very industry that was an early advocate. Hungry for free new spectrum and rightfully concerned about losing market share to cable, DBS and pre-recorded media, broadcasters fought for DTV spectrum and even (reluctantly) committed to a timetable for a DTV signal.

But now that they are being asked to meet the timetable they agreed upon, some broadcasters are clamoring for delay. Even as responsible broadcasters like CBS move forward aggressively with DTV, others are badmouthing this exciting new technology at the same time they hold tightly on to their free new spectrum.

And, unfortunately, some broadcasters believe delay is in their interest, as they don't have to return spectrum until 85 percent of viewers in their area have digital reception.

Fortunately, the cable and satellite industries hunger for digital television. They see new revenue opportunities and quality programming with enthusiastic consumers.

Just around the corner is pre-recorded DTV programming and, soon, broadband will allow Internet, and maybe even videogame, DTV.

In this issue of DTV Guide, we have the latest report on DTV programming from cable, satellite and broadcast sources – and news about some broadcasters' plans to deliver data and other services to consumers. You'll find the latest market-by-market list of local DTV stations and detailed charts of available DTV products.

Introductory sales have been great, despite limited programming availability. Consumers are excited about the technology and many are buying it. As you'll read in three testimonials from DTV retailers, those who've invested in the technology are having no regrets.

Our industry's challenge is to deliver on DTV and build consumer confidence even while others are pushing for delay. The good news is that DTV offers such a phenomenally improved experience that it will win on its own in the marketplace. Do your best to keep the pressure on local broadcasters—and reward those that are helping, rather than hurting, our momentum.

Thanks to **TWICE** and Broadcasting & Cable for partnering with CEA on this issue of DTV Guide, and helping us keep the industry up-to-date on the digital transition. ■

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KEN CRANE'S: L.A.'S DTV DESTINATION



*Tom Campbell
Ken Crane's Home
Entertainment City, Inc.*

LOS ANGELES — Rarely in the 51 years that Ken Crane has sold consumer electronics products has a new category produced the excitement level that we are seeing from high-definition television today.

Thanks to the entrepreneurial spirit of company founder Ken Crane and his son, vice president Casey Crane, Ken Crane's Home Entertainment City has become a leading destination point for digital televisions in the Greater Los Angeles area.

Although it may seem like a no-brainer now, the decision to take on the first digital TVs two years ago had some risks. In 1998, no one really knew for sure that the public would consider investing in a new class of digital televisions costing hundreds or thousands of dollars more than comparably sized analog models, which continue to sell very well.

If we brought these new sets onto the sales floor, would they confuse and confound the public

so much that the current healthy TV business would be adversely impacted? Ken Crane bit the bullet, and brought in a small HDTV assortment to test the waters. He was amazed at the reaction.

I'm pleased to report the benefits of HDTV have far outweighed the risks. Consumers are showing a lot of interest in HDTV. The feedback has been very positive, and we have seen double-digit sales increases in digital-upgradeable televisions every month since the first month we began selling them.

Some 68 percent of our overall television business today is done in HDTV upgradeable models, and 20% of the customers who bought one of those models also purchased a digital set-top decoder box to watch off-air digital broadcasts (and sometimes digital satellite programs in high-definition). These decoders make the HDTV upgradeable TVs a fully functioning digital TV system and give broadcasters a living, breathing audience.

We carry a broad selection of

digital televisions from Mitsubishi, Panasonic, Pioneer and Sony, and we carry the set-top decoders from Mitsubishi, Panasonic and RCA.

In addition, HDTV has created a renewed interest in the total home theater concept. Ken Crane has enjoyed increased business in home theater audio products, DVD players and other high-performance system components, as a result of consumers' seeing the breath-taking clarity of this new class of big-screen television combined with a Dolby Digital surround sound presentation. In fact, I believe the key reason a customer buys an HDTV upgradeable set today is to watch standard DVDs in enhanced picture quality, often with a filled widescreen picture.

I also believe we have been helped by the availability of digital television broadcasts in our market. There are seven broadcasters transmitting digital over-the-air programming in the Greater Los Angeles area. This generates con-

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DIGITAL TELEVISION: A TECHNOLOGY YOU CAN BANK ON



*By Noah Herschman,
Vice President, Marketing Tweeter
Home Entertainment Group*

Even though Digital Television feels like it is currently waylaid in some sort of purgatory, we have already experienced the first ripple of the impending storm. Since the advent of HDTV-compatible (HDTV-C) sets two years ago, Tweeter Home Entertainment Group's projection television sales have more than doubled. Currently HDTV-compatible sets account for more than 60% of our projection television (PJTV) sets in units and over 80 percent in dollars.

This is not inconceivable given the fact that the early adopter is one of our primary customers. What has been a pleasant surprise is that the lower tiers of customers on the enthusiasm curve are also embracing high-definition technology.

Instead of seeing sales of HDTV-compatible sets slow down after the initial early adopter pipeline was filled, sales have increased and we're selling more of these sets than ever.

Perhaps this is due to the fact that these sets, with their exemplary displays and line-doubling

technology deliver by far, the best analog pictures we have ever seen. But the amazing sales growth is also due to today's customer's savvy with regard to the future.

Our customers (whom we maintain are smarter than our competitors) have been burned by the obsolescence flame before. They have purchased 64,000 computers and DIVX players only to see those products replaced by faster and better-performing ones. They understand the value of upward-compatibility and are willing to pay a premium to insure that their new purchase has legs. It is actually an easy sell if you think about it: you can pay \$1,500 for an analog PJTV or \$2,500 for an HDTV-compatible one.

The question of programming is now at the forefront of the debate. But we have confidence that programmers who are committed to high quality will prevail in this arena. We applaud HBO and DirecTV for taking a stand and delivering premium programming and pay-per-view movies in high-definition, and we look forward to being able to demonstrate this programming in our stores. (HBO: Please make sure you run some G and PG-rated movies during regular business hours).

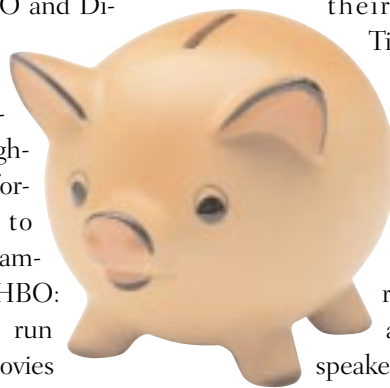
The future looks bright for A/V retailers who have become comfortable selling DTV. Although all of the forces still need to be

aligned, especially where programming is concerned, look at the ramifications: When the CD player was introducing in the mid-80's, hi-fi stores saw an unprecedented boom in sales. Not only did we sell the CD player itself, but because of its superior fidelity (mid-fi anyway), it initiated a domino effect.

Customers felt the urge to upgrade their entire system – receiver, speakers, cassette recorder, speaker wire – everything had to be “digital-ready.” Multiply what happened in the 80's by a factor of five and you'll come close to the phenomenon we are about to witness in '00.

A 1080i HDTV picture is six times more detailed than an analog one. And Dolby Digital is immeasurably better than the sound from a regular TV set. The upgrade possibilities are astounding.

Besides the TV itself, customers are compelled to replace their VCR with TiVo or Replay; and purchase a progressive scan or HD D V D player, a surround sound receiver, and at least six speakers. We're up to almost \$2,000 over and above the price of the TV! Do the math, if now is not the time to be a hi-fi salesman, I don't know when will be. ■



BEST BUY TAKES DTV LEAD

By Lee Simonson, Best Buy

MINNEAPOLIS, Minn. — As a company, Best Buy is very pleased with the progress of our digital television business.

According to NPD Intellect market research data, we hold the No. 1 market share position in digital television sales. That means, we are selling digital televisions better than anyone in the country. This can be attributed to the organization's commitment to building the category from its inception.

In November 1998, Best Buy became the first retail chain to advertise the digital television system and products nationally. This has helped to spark the imaginations of American consumers, drawing

thousands of people to local Best Buy stores to see high definition television up close. Today, we struggle to get enough inventory to meet the demands of our customers.

Although the early activity was encouraging, it wasn't until last April that digital television sales became what I would call truly robust. That was when Sony began deliveries of its current generation of Hi-Scan digital-compatible rear projection televisions. This was an important milestone for two reasons, it alleviated the product shortage conditions that have limited the sales volume of digital compatible televisions for Best Buy and the consumer electronics industry. Previously, the availability



of digital-capable televisions and set-top digital decoder boxes had been so limited that digital sets weren't shown in some locations.

The second reason: Some say Sony's newly expanded assortment also validated the category, and encouraged other manufacturers to market their digital television lines more aggressively. The industry has done a wonderful job in bringing the price of digital-ready 4:3 (conventional square-screen) tele-

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RETAILERS

RETAILERS CARRYING DIGITAL TELEVISION PRODUCTS

ATLANTA MARKET

Audio Forest	Atlanta
Laserdisc Enterprises	Smyrna
Georgia Music	Macon
Best Buy	Athens
Circuit City	Atlanta
Hi-Fi Buys	Atlanta
Mundy's Audio Video	Gainesville
Roberts	Decatur
Sears	Atlanta

BALTIMORE MARKET

Best Buy	Columbia
Bryn Mawr Stereo	Glen Burnie
The Big Screen Store	Lutherville
Circuit City	Catonsville
Gramophone	Lutherville
Sears	Baltimore
White Marsh TV Sales	White Marsh

BOSTON MARKET

Best Buy	Cambridge
Cambridge Soundworks	Cambridge
Circuit City	Somerville
Doyons TV & Appliance	Reading
Goodwin's High End Audio	Waltham
Huntington TV	Chestnut Hill
Natural Sound	Framingham
Percy of Worcester	Worcester
Sears	Cambridge
Stereo Shop	Boston
Tweeter	Boston

CHARLOTTE MARKET

Audio Video Systems	Charlotte
Best Buy	Charlotte
Circuit City	Charlotte
Queen City TV	Charlotte
Sound Systems	Charlotte
Stereo Showcase	Charlotte

CHICAGO MARKET

Abt Electronics	Morton Grove
Absolute Audio Video	Rockford
Barretts Audio-Video	Elmhurst
Best Buy	Addison
Columbia Audio Video	Highland Park
Dan's TV	Aurora
Douglas TV	Chicago
Plas Appl. & Electronics	Oak Lawn
Quintessence	Naperville
Sears	Chicago
Steger TV	So. Chicago
Trage Bros	Forest Park
United Audio Centers	Chicago

CINCINNATI MARKET

Alamo Electronics	Cincinnati
Best Buy	Cincinnati
Circuit City	Cincinnati
Marvine's TV	Franklin
Roberts	Springdale
Sears	Cincinnati

CLEVELAND MARKET

Audio Craft	Cleveland
B&B Appliance	Middleburg Hts.
Best Buy	Cleveland
Circuit City	Cleveland
Mentor TV	Cleveland
Sears	Cleveland

DALLAS/FORT WORTH MARKET

Best Buy	Dallas
Circuit City	Plano
Consumer Electronics	Arlington
Dillard's	Dallas
Ed Kellum & Son	Dallas
Fry's	Dallas
Hillcrest High Fidelity	Dallas
Home Entertainment	Dallas
Goodman Radio	Dallas

Jim Gunter TV & VCR	Fort Worth
Home Theater Store	Dallas
Kyrstal Clear Audio Video	Dallas
Marvin Electronics	Fort Worth
Sears	Dallas
Sound Idea	Arlington
Star Power Home Theater	Dallas
Stereo East	Dallas

DENVER MARKET

Best Buy	Denver
Circuit City	Denver

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Kacey's Fine Furniture	Denver
Laserland	Denver
Listen Up	Denver
Soundtrack	Denver
Valas TV	Denver

DETROIT MARKET

ABC Warehouse	Dearborn
AMJ Electronics	Gross Pointe
Alma's Hi Fi Stereo	Dearborn
Audio Video Alternatives	Royal Oak
Bekins Audio Video	Grand Rapids
Best Buy	Dearborn
Circuit City	Detroit
The Gramophone	Birmingham
Listening Room	Saginaw
NBS	Clair Shores
Northern Television & Video	Royal Oak
Pecar Electronics	Detroit
Sears	Lakeside

HARTFORD/NEW HAVEN MARKET

Cambridge Soundworks	Farmington
Circuit City	Manchester
Sear's	Hartford
The Stereo Shop	Hartford
Tweeter	Newington

HOUSTON MARKET

All-Star Audio Video	Houston
Audio Concepts	Houston
Best Buy	Houston
Circuit City	Houston
Conn Appliance	Houston
Dillard's	Houston
Home Entertainment	Houston
Sears	Houston
Videoland	Houston

INDIANAPOLIS MARKET

Best Buy	Indianapolis
Circuit City	Indianapolis
H.H. Gregg	Indianapolis
Ovation Audio-Video	Indianapolis
Sears	Indianapolis
Sound Pro	Indianapolis

LOS ANGELES MARKET

Ahead Stereo	Los Angeles
Allen's TV	Oxnard
Ametron Audio Video	Los Angeles
Audio/Video City	Los Angeles
Audio Video Today	Westminster
Azusa Sales Audio Video	Azusa
Best Buy	Los Angeles
Caston's TV & Appliance	Newhall
Circuit City	Glendale
Cosmos	Los Angeles
Dearden's	Los Angeles
Fry's	Los Angeles
Hollytron	Los Angeles
Howard's Appliances	Laguna Hills
Good Guys!	Glendale
Ken Cranes Home Ent.	Hawthorne
LA Audio Video	Los Angeles
Marconi Radio TV Hi Fi	Glendale
Paul's TV	Los Angeles
Sears	Los Angeles
Shelley's Stereo	Woodland
Star Club	Los Angeles
Supervision	Los Angeles
Wilshire TV & Stereo	Thousand Oaks

MIAMI MARKET

Appliance TV Depot	Gables
Best Buy	Miami
Brandsmart	Miami
Circuit City	Miami
Sears	Miami
Sound Advice	Gables
Sound Components	Coral Gables

MINNEAPOLIS/ ST. PAUL MARKET

Audio King	Minneapolis
Audio Perfection	Minneapolis
Best Buy	Richfield

Circuit City	Bloomington
Edina Audio King	Edina
Elite Audio	Eau Claire
Hi-Fi Sound Electronics	Minneapolis
Robert Paul Co.	St. Paul
Sears	St. Paul
Sound Center	Minneapolis

NEW YORK MARKET

6th Avenue Electronics	Springfield
AVI	Little Falls
Best Buy	Princeton
Capital Audio Electric	New York
Circuit City	Manhattan
Freehold Stereo & Video	Freehold
Harvey Electronics	New York
J&R Music World	New York
Lyric Hi-Fi	New York
MTV Electronics	Mount Vernon
Nationwide	Edison
Nobody Beats the Wiz	New York
P.C. Richard	New York
Robson's Audio Video	Paramus
Sears	Jersey City
Sound by Singer	New York
Stereo Effects	Great Neck
Stereo Exchange	New York
Stuarts Audio	Westfield
Tops	Edison
T.R. Technologies	New York
Ultra Hi-Fi	Flushing
Woodbridge Stereo Video	Woodbridge

ORLANDO MARKET

Absolute Sound	Orlando
Best Buy	Orlando
Circuit City	Orlando
Sears	Orlando
Sound Advice	Orlando

PHILADELPHIA MARKET

Audio Images	Whitehall
American Appliance	Montgomery
Audio Video Concepts	Cherry Hill
Best Buy	Oxford Valley
Bryn Mawr Stereo	Bryn Mawr
Circuit City	Philadelphia
Danby Radio	Ardmore
Hi Fi House	Broomall
Hi Fi Sales	Cherry Hill
Sears	Philadelphia
Smith & Son	Medford
Soundex High End A/V	Willow Grove

PHOENIX MARKET

Best Buy	Phoenix
Buzz Jensen's Sound Advice	Phoenix
Circuit City	Phoenix
Dennis Sage Home Ent.	Phoenix
Denny's TV & Appliance	Sun City
Fry's	Tempe
Jerry's A/V	Phoenix
Hi Fi Sales	Mesa
Romano's TV & Appliance	Scottsdale
Sear's	Scottsdale
Sound Advice	Phoenix
Sounds Like Music	Phoenix
Walt's TV & Appliance	Tempe

PITTSBURGH MARKET

Best Buy	Pittsburgh
Circuit City	Pittsburgh
Daniel's	Pittsburgh
Don's Appliance	Pittsburgh
Pat's Stereo Center	Greensburg
Stereo Outlet	Washington
Stereo Shop	Greensburg
Triangle Radio & Television	Pittsburgh
Voss TV & Appliance	Pittsburgh

PORTLAND MARKET

Captain Video	Lake Oswego
Chelsea Audio Video	Portland
Circuit City	Tigert
Fry's	Portland
The Good Guys	Portland
Home Video Library	Portland
Magnolia Hi Fi	Beaverton
Northwest Home Electronics	Lake Oswego
Standard TV & Appliance	Portland

RALEIGH/DURHAM MARKET

Best Buy	Raleigh
Circuit City	Raleigh
Garner TV & Appliance	Garner
Kelly's	Fayetteville
Now Audio Video	Raleigh
Sears	Raleigh

SACRAMENTO MARKET

Circuit City	Sacramento
Good Guys	Citrus Heights

SAN DIEGO MARKET

Big Screen City	La Jolla
Circuit City	San Diego
Convoy Audio Video	San Diego
Dow Stereo	San Diego
Encinitas Television	San Diego
Fry's	San Diego
Tommy's TV	San Diego
The Good Guys!	San Diego
Main Street Electronics	San Diego
Soundhaus	El Cajon
Stereo Unlimited	San Diego

SAN FRANCISCO MARKET

Audio Excellence	San Francisco
Anderson TV & Stereo	Redwood City
Best Buy	San Francisco
Century Stereo	San Jose
Circuit City	San Francisco
Eber Electronic Supply	San Francisco
Fry's	San Jose
The Good Guys!	San Francisco
Laser Land Home Video	San Jose
Laser City	San Francisco
Performance Audio	San Francisco
Sears	Santa Rosa
Sound Systems	Santa Clara
Stereo Plus	San Francisco
Telecenter Appliances	San Mateo
Wong's Audio Visual	San Francisco

SEATTLE MARKET

Circuit City	Seattle
Definitive Audio	Seattle
Dresco	Seattle
The Good Guys!	Seattle
Magnolia Hi-Fi	Seattle
Sears	Seattle

ST. LOUIS MARKET

Best Buy	St. Louis
Circuit City	St. Louis
Cohen Appliance & TV	St. Louis
Hi-Fi Fo-Fum	St. Louis
Sears	St. Louis
The Sound Room	St. Louis

TAMPA MARKET

Best Buy	Tampa
Circuit City	Tampa
Miami Audio Video	Tampa
Rex TV	St. Pete.
Roberts	Tampa
Sears	Tampa
Sound Advice	Tampa

WASHINGTON D.C. MARKET

Audio Video Interiors	Reisterstove
Belmont TV	Silver Springs
Best Buy	Reston
Big Screen Store	Rockville
Circuit City	Rockville
Domes	Chesapeake
Goryo Electronics	Lorton
Graffiti Audio Video	Washington
Myer-Emco	Washington
Multimedia Solutions	Crownsville
Murrells Television	Washington
Paul's Audio Video Center	Vienna
Pro Video	Washington
Sears	Alexandria
Soundworks Audio/Video	Kensington
Theater Vision	Rockville

BROADCASTERS PROMISE MORE HDTV TO COME

By Glen Dickson

If 1999 was the year HDTV rocketed to prominence, the year 2000 has seen the new medium remain in a holding pattern. Unlike last spring, when CBS and ABC announced innovative subsidy deals with TV set makers to support weekly HDTV programming, this year has passed with no significant HDTV programming announcements.

To be fair, 2000 has seen some new HDTV programs, mainly in the form of sporting events. ABC produced the NHL All-Star Game in HDTV in February, after showing *Monday Night Football* and the Super Bowl in hi-def with help from Panasonic. In addition to its Mitsubishi-sponsored prime-time fare, CBS showed high-definition broadcasts of the NCAA Men's Basketball Tournament and the Masters golf tournament in April (the NCAA Final Four was subsidized by Thomson Consumer Electronics, while the Masters was indirectly supported by Sony, which

supplied gear to tournament host Augusta National). NBC showed the NBA All-Star Game in HDTV in February, its first hi-def sports production.

But neither Mitsubishi nor Panasonic have agreed to renew their respective subsidy deals with CBS and ABC, and so far neither network has announced its HDTV programming

lineup for this fall. Mitsubishi has indicated it will focus on sports and other event programming with its sponsorship dollars, while Panasonic has sold the \$6 million 720P production truck it built for *Monday Night Football* to The Ackerley Group, a Seattle-based station owner. NBC, which has been showing *The Tonight With Jay Leno* in HDTV since May 1999, hasn't announced any new HDTV plans, although the network has previously indicated it would broadcast *Titanic* in 1080i this year.

CBS continues to broadcast 15 hours of HDTV a week and is still pursuing HDTV subsidy deals for the upcoming season, says spokesman Dana McClintock. "We have a similar model on the table with various manufacturers, and we do expect announcements in the coming weeks," says McClintock, who says that CBS is considering more big-event HDTV programming (industry sources suggest it will be NFL football). He says it's "unlikely" that CBS would air HDTV programming without it being subsidized.

Mitsubishi director of marketing Bob Perry emphasizes that his company will continue to work with CBS. "We have not severed our relationship with CBS," says Perry. "However, we will not renew our exclusive primetime deal. We're just changing our strategy."

After being unable to close another agreement with Panasonic to

sponsor *Monday Night Football* in hi-def, ABC is considering other options for this fall. But they don't include *Monday Night Football*.

ABC decided in late June to call off talks with The Ackerley Group about renting its 720P production truck for the 2000 season of MNF. Since there aren't any other 720p trucks in existence, says ABC President of Broadcast Operations and Engineering Preston Davis, "at this point, it does not look like we will be doing MNF in HDTV." He adds that ABC will increase the number of HDTV movies it broadcasts. The network is also considering some HDTV primetime series for the fall, which may be subsidized by consumer electronics manufacturers.

"I don't think that will drive the decision one way or the other, but we are in discussions with c.e. companies," says Davis.

NBC will continue to broadcast *Tonight* and occasional special events in HDTV, says Steve Pair, vice president of engineering for NBC's station group. The costs of delivering a separate high-definition satellite feed have made doing more prohibitive, he says.

"We're in the process of planning and implementing a new delivery system" says Pair. "Right now, we have to book special [satellite] time for HDTV. We're currently looking at a year or two rollout to digital delivery, then we'll do as much HDTV as we can."

While Fox doesn't broadcast any

DTV Programming

HDTV, the network has been pumping out an all-digital primetime feed since November 1998, as well as the 480p widescreen Fox News Sunday. All of Fox's primetime film content is produced in 480p, while reality-based programming shot on videotape is converted from 480i to 480p for transmission. About three hours of Fox's primetime schedule are in widescreen, including shows such as *Ally McBeal* and *The X-Files*, and the network expects to broadcast up to eight hours of widescreen per week this fall.

Rounding out the commercial networks, neither UPN nor The WB has announced any HDTV plans. PBS, on the other hand, is showing several HDTV and widescreen SDTV productions each month in primetime. The rest of the time, the public television network is using its DTV spectrum to multicast four SDTV streams, including east and West Coast feeds of its primary programming, a children's channel, and adult education channel. PBS has also experimented with enhanced broadcasting, transmitting associated data with children's programming and documentaries.

John Tollefson, Chief Technology Officer for PBS, says the network will do more HDTV when it becomes cheaper to produce or when producers see a demand.

"The basic strategy hasn't changed any, but we have taken a more realistic view of what can be done," says Tollefson. "There's not a whole lot of high-definition sets at home. But we still look forward to the day when most of the schedule will be in high-definition. Clearly, there are some events that benefit

greatly from [it]."

Among cable networks, MSG Network and HBO are the clear leaders in HDTV. MSG, a regional sports network primarily carried on cable systems in the New York area, is currently producing its second season of New York Yankees and Mets baseball in 1080i HDTV (in the winter, it produces the New

York Knicks and Rangers and New Jersey Nets and Devils in HDTV). HBO has been airing HDTV movies since March 1999, and is being carried by cable operators Time Warner and Cablevision as well as DirecTV and EchoStar.

"We're chugging away, and there have been no changes in our course at all," says Bob Zitter, senior vice president of technology operations for HBO. Zitter adds that HBO has been showing over 60 percent of its movie titles in HDTV for the past few months, and offered its first HD title with 5.1-channel Dolby Digital sound in June.

HBO hasn't started to produce any of its original programming such as *The Sopranos* in HDTV, and Zitter doesn't see that changing any time soon. Zitter says the major difficulty is that HDTV post-production equipment necessary for original production hasn't been widely avail-

able. But with new 1080p/24-frame gear being rapidly accepted in Hollywood, says Zitter, "12 months from now there could be a very different answer to that."

Zitter acknowledges that HDTV set penetration has been slow, primarily due to interface issues with cable set-tops and the relatively limited flow of programming. But he doesn't see any major stumbling blocks to the new format.

"We knew when we started we were going way ahead of the curve in terms of accessibility to our subscribers," says Zitter. He expects more major MSOs to start distributing HBO's hi-def service in late 2000 or early 2001.

EchoStar and Time Warner are also carrying HDTV movies from Showtime, which launched its hi-def service in January. Showtime is showing close to 50 percent of its primetime movies in HDTV, says Glenn Oakley, Showtime senior vice president of corporate strategy. The network is also augmenting that HDTV content by procuring widescreen SDTV masters (with Dolby Digital sound) of movies and original programming and upconverting them to 1080i for transmission. That means that about 75 percent of Showtime's programming is available in widescreen DTV.

"We've been very explicit in telling people it's not high-definition," says Oakley of the widescreen SDTV programming. "But those people that have seen it, say it's virtually indistinguishable." ■

Glen Dickson is associate editor of Broadcasting & Cable, a publication of Cahners Business

DTV MARKET OUTLOOK

By Todd Thibodeaux

Sales of DTV display devices (with or without an integrated decoder) are expected to total 425,000 units in 2000, up more than 250 percent from total display shipments in 1999. Rear projection displays should account for roughly 75 percent of all sales this year with the remainder coming from direct-view devices. The majority (57 percent) of projection units will have screens 55" and larger (including sets with a 16:9 or 4:3 aspect ratio). The DTV projection display market is decidedly more "big screen" than its analog sister market. Among analog projection displays, only 30 percent of sets sold in 2000 will have screens 55" and over. Virtually all DTV projection displays to be sold in 2000 will be capable of displaying programming at HDTV 720p or 1080i, up from only about 50 percent for sales made during 1999. The balance of 4:3 projection displays and 16:9 models will tip in favor of the wider screen format by the end of 2000. A trend likely to continue to gain more momentum as DVD owners look to get the most from their home theater systems.

Going forward, the DTV market really has three paths it can follow depending on how programming and interactive services are deployed. If the market sees stepped up efforts from broadcast-

ers to accelerate the rollout of digital programming, and two-way interactive data services finally get on track, the prospects for digital TV display penetration are greatly enhanced. Compared to a middle of the road forecast, aggressive efforts should push display penetration to 50 percent by 2006 with an installed base of more than 56 million displays. If the market continues to be characterized by trepidation and a lack of commitment to programming, digital TV will not reach its true potential – and consumers may never have full access to this extraordinary technology. Under such a scenario, penetration will rise slowly to only 15 percent and the installed base will total less than 17 million.

The market for digital TV products is complicated, but full of huge potential. Many forces including products, programming, standards, services, and others, are blending together in sometimes odd ways paving the road traveled so far. The prize at the end is worth the trip for consumers and all the industries involved in the transition. ■

Todd Thibodeaux is CEA's Vice President of Market Research.

Product Profiles:



Faroudja offers broadcasters the chance broadcast digital TV formats from an video source material in its Digital Format Translator (above). The device will convert any video source into 480p, 720p or 1080i formats, while simultaneously converting the aspect ratio of the input image into the 16:9 ratio of the new cinema-style TV monitors.



Runoco's VHD-4404 Ultra video processor (\$19,995) has been designed to transform today's analog signal sources into a virtual high definition image. The unit include line doubling/quadrupling, and 3:2 pull down detection clean conversion of film-based video source material.

DTV Products

The following CEA/TWICE charts contain the most comprehensive source of information on DTV products available in the market.

All of the DTV products listed fall into one of the following categories:

- DTV and HDTV monitors, which, with the addition of a digital set-top box, offer a complete DTV system,
- integrated high-definition sets that include a digital receiver and display; or
- digital set-top boxes designed to work with high-definition and digital displays, or current analog displays.

1080i and/or 720p-Capable Monitors (15.7kHz-33.75kHz or better)

Brand	Model	Display Type	Screen Size	HDTV Scan Rate Display Capability	On-Board Line Doubling/Scaling?	Number of NTSC Tuners	Interface for DTV Tuner/Decoder	Available	Price
Cinevision	1050	7" CRT Front PTV	Variable	720p, 1080i	No	0	VGA 15-pin D-sub	Now	\$20,000.00
Dwin	HD500	7" CRT Front PTV	Variable	720p, 1080i (1280x1024)	No, optional line doubler, multiplier available	0	RGB H&V/C Sync via BNC	Now	\$12,000.00
Dwin	HD700	7" CRT Front PTV	Variable	720p, 1080i	No, optional line doubler, multiplier available	0	RGB H&V/C Sync via BNC	Now	\$12,500.00
Faroudja	RP-5800	CRT Rear PTV	58W" 16:9	720p, 1080i	Yes	0	RGBS-H&V via VGA 15-pin D-sub	Now	\$35,000.
Fujitsu	PDS-4221	Plasma Panel (1024x1024)	42W" 16:9	720p, 1080i	Scaler	0	RGB-H&V BNC, Component BNC, RGB 15-pin D SUB	Now	\$15,995.00
Fujitsu Brushed Silver	PDS-4222	Plasma Panel (1024x1024)	42W" 16:9	720p, 1080i	Scaler	0	RGB-H&V BNC, Component BNC, RGB 15-pin D SUB	Now	\$15,995.00
Fujitsu	PDS-4209	Plasma Panel (852x480)	42W" 16:9	720p, 1080i	Scaler	0	RGB-H&V BNC, Component BNC, RGB 15-pin D SUB	July	\$7,999.00
Fujitsu	PDS-4214	Plasma Panel (852x480)	42W" 16:9	720p, 1080i	Scaler	0	RGB-H&V BNC, Component BNC, RGB 15-pin D SUB	July	\$9,999.00
Hitachi	55DMX01W	DLP Rear PTV	55W" 16:9	720p	Yes	2	RGB-H&V BNC, Component BNC, RGB 15-pin D SUB	October	\$9,999.00
Hitachi	61SWX01W	7" CRT Rear PTV	61W" 16:9	1080i	Yes	2	(2) HD Component	Sept.	\$4,999.00
Hitachi	61SDX01B	7" CRT Rear PTV	61" 4:3	1080i	Yes	2	(2) HD Component	July	\$3,799.00
Hitachi	53SWX01W	7" CRT Rear PTV	53" 16:9	1080i	Yes	2	(2) HD Component	Sept.	\$3,999.00
Hitachi	53SDX01B	7" CRT Rear PTV	53" 4:3	1080i	Yes	2	(2) HD Component	June	\$3,299.00
Hitachi	53FDX01B	7" CRT Rear PTV	53" 4:3	1080i	Yes	2	(2) HD Component	July	\$2,699.00
Hitachi	43FDX01B Tabletop	7" CRT Rear PTV	43" 4:3	1080i	Yes	2	(2) HD Component	August	\$2,199.00
Hitachi	36SDX88B	Direct View	36" 4:3	1080i	Yes	1	HD Component, 15-pin D-sub	Now	\$2,199.00
Hitachi	36SDX01S	Direct View	36" 4:3	1080i	Yes	2	HD Component	July	\$2,199.00
Hitachi	53SDX89B	7" CRT Rear PTV	53" 4:3	1080i	Yes	2	HD Component	Now	\$3,299.00
Hitachi	60SDX88B	7" CRT Rear PTV	60" 4:3	1080i	Yes	2	HD Component	Now	\$3,799.00
JVC	AV-61S901 D'Alila	D-ILA Hologram Rear PTV	61W" 16:9	1080i (1.32 mil. pixels)	Yes line doubles NTSC	2	(2) HD Component	June	\$6,999.00
Konka	HR3093U	Flat-Faced Direct View	30W" 16:9	1080i	Yes	1	RGB High Density 15-way D-type socket	Sept.	\$2,499.00
Konka	HR3289U	Flat-Faced Direct View	32" 4:3	1080i	Yes	1	RGB High Density 15-way D-type socket	2001	TBA
Konka	HD5098U	Rear Projection	50" 16:9	1080i	Yes	1	RGB High Density 15-way D-type socket	April	TBA
Madrigal	MP-9	9" CRT Front PTV	Variable	1080i, 720p	No	0	RGB H&V sync,	Now	\$60,000.00
Madrigal	MP-8	8" CRT Front PTV	Variable	1080i, 720p	No	0	RGB H&V sync,	Now	\$45,000.00
Marantz	VP-8000	DLP Front PTV	Variable	1080i	Yes	0	HD Component	Now	\$9,999.00
Mitsubishi Platinum Series	WT-46805	7" CRT Rear PTV	46W" 16:9	1080i	Yes	2	HD Component, RGB via VGA	Now	\$3,799
Mitsubishi Platinum Series	VS-50805	7" CRT Rear PTV	50" 4:3	1080i	Yes	2	HD Component, RGB via VGA	Now	\$3,999
Mitsubishi Platinum Series	WS-55805	7" CRT Rear PTV	55W" 16:9	1080i	Yes	2	HD Component, RGB via VGA	Now	\$4,499
Mitsubishi Platinum Series	VS-60805	7" CRT Rear PTV	60" 4:3	1080i	Yes	2	HD Component, RGB via VGA	Now	\$4,499
Mitsubishi Platinum Series	VS-80803	9" CRT Rear PTV	80" 4:3	1080i	Yes	2	Proprietary RGB	Now	\$9,999
Mitsubishi Diamond Series	WS-55905	7" CRT Rear PTV	55W" 16:9	1080i	Yes	2	HD Component, RGB via VGA	Now	\$5,499
Mitsubishi Diamond Series	WS-65905	7" CRT Rear PTV	65W" 16:9	1080i	Yes	2	HD Component, RGB via VGA	Now	\$6,999
Mitsubishi Diamond Series	WS-73905	9" CRT Rear PTV	73W" 16:9	1080i	Yes	2	HD Component, RGB via VGA	Now	\$9,999

Courtesy of TWICE

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1080i and/or 720p-Capable Monitors (15.7kHz-33.75kHz or better) — continued from page 16

Brand	Model	Display Type	Screen Size	HDTV Scan Rate Display Capability	On-Board Line Doubling/Scaling?	Number of NTSC Tuners	Interface for DTV Tuner/Decoder	Available	Price
Mitsubishi Platinum Series	WT-46807	7" CRT Rear PTV	46W" 16:9	1080i	Yes	2	HD Component	September	\$3,499.00
Mitsubishi Platinum Series	WS-55807	7" CRT Rear PTV	55W" 16:9	1080i	Yes	2	HD Component	September	\$3,999.00
Mitsubishi Platinum Series	WS-65807	7" CRT Rear PTV	60W" 16:9	1080i	Yes	2	HD Component	August	\$4,499.00
Mitsubishi Platinum-Plus Series	WS-55857	7" CRT Rear PTV	55W" 16:9	1080i	Yes	2	HD Component	August	\$4,499.00
Mitsubishi Platinum-Plus Series	WS-65857	7" CRT Rear PTV	65W" 16:9	1080i	Yes	2	HD Component	September	\$4,999.00
Mitsubishi Diamond Series	WS-55907	7" CRT Rear PTV	55W" 16:9	1080i	Yes	2	HD Component	September	\$4,999.00
Mitsubishi Diamond Series	WS-65907	7" CRT Rear PTV	65W" 16:9	1080i	Yes	2	HD Component	September	\$5,499.00
Mitsubishi Diamond Series	WS-73907	9" CRT Rear PTV	73W" 16:9	1080i	Yes	2	HD Component	October	\$9,999.00
Mitsubishi DLP	WD-65000	1-chip DLP Rear PTV	65W" 16:9	1080i	Yes	2	HD Component	December	\$9,999.00
NetTV	DTV29X	Direct View	27" 4:3	1080i, 720p	No	0	(2) VGA 15-pin D-sub	Now	\$1,299.00
NetTV	DTV29XC	Direct View	27" 4:3	1080i, 720p	No	0	(2) VGA 15-pin D-sub HD Component	Now	\$1,899.00
NetTV	DTV34X	Flat Faced Direct View	32" 4:3	1080i, 720p	No	0	(2) VGA 15-pin D-sub HD Component	Now	\$1,849.00
NetTV	DTV34XC	Flat Faced Direct View	32" 4:3	1080i, 720p	No	0	(2) VGA 15-pin D-sub HD Component	Now	\$2,599.00
NetTV	DTV36XW	Flat Faced Direct View	34W" 16:9	1080i, 720p	No	0	(2) VGA 15-pin D-sub HD Component	Now	\$4,599.00
NetTV	DTV42WP	Plasma Display Panel	40W" 16:9	1080i, 720p	No	0	RGB, HD Component	Now	\$11,999.00
Panasonic	CT-34WX50 Tau	Flat-Faced Direct View	34W" 16:9	1080i	Yes	1	HD Component	May	\$5,999.00
Panasonic	CT-32HX40 Tau	Flat-Faced Direct View	32" 4:3	1080i	Yes	2	HD Component	Oct/Nov.	\$1,999.00
Panasonic	CT-36HX40 Tau	Flat-Faced Direct View	36" 4:3	1080i	Yes	2	HD Component	Nov./Dec.	\$2,299.00
Panasonic	PT-52DL10	DLP Rear PTV	52" 16:9	720p (1280x720)	Yes	1	HD Component	March	\$12,000.00
Panasonic	PT-56WXP95	7" CRT Rear PTV	56W" 16:9	1080i, 720p	Yes	2	HD Component	Now	\$5,499.95
Panasonic	PT-65WX50	7" CRT Rear PTV	65W" 16:9	1080i, 720p	Yes	2	HD Component	March	TBA
Panasonic	PT-51HX40	7" CRT Rear PTV	51W" 4:3	1080i	Yes	2	HD Component	June/July	\$2,599.00
Panasonic	PT-56HX40	7" CRT Rear PTV	56W" 4:3	1080i	Yes	2	HD Component	June/July	\$2,799.00
Panasonic	PT-61HX40	7" CRT Rear PTV	61W" 4:3	1080i	Yes	2	HD Component	June/July	\$3,099.00
Philips	60PP9701	7" CRT Rear PTV	60W"	1080i	Yes	2	HD Component	Summer	\$5,299.00
Philips	55PP9701	7" CRT Rear PTV	55W" 16:9	1080i	Yes	2	HD Component, VGA 15-pin D-sub	Summer	\$4,799.00
Philips	60PP9601	7" CRT Rear PTV	60" 4:3	1080i	Yes	2	HD Component	Summer	\$3,799.00
Philips	30PW9815	Direct View	30W" 16:9	1080i	Yes	2	HD Component	July	\$3,000.00
Philips	34PW9815	Direct View	34W" 16:9	1080i	Yes	2	HD Component	July	\$3,499.00
Pioneer	PDP-505HD	Plasma Panel	50W" 16:9	720p, 1080i (1280x768)	Yes	0	RGB, HD Component	Now	\$20,000.00
Pioneer	SD-641HD5	7" CRT Rear PTV	64W" 16:9	1080i	Yes	2	HD Component	Now	\$6,999.00
Pioneer	SD-532HD5	7" CRT Rear PTV	53W" 16:9	1080i	Yes	2	2) VGA 15-pin D-sub HD Component	Now	\$4,499.00
Pioneer	SD-582HD5	7" CRT Rear PTV	58W" 16:9	1080i	Yes	2	2) VGA 15-pin D-sub HD Component	Now	\$5,499.00
Pioneer Elite	PRO-510HD	7" CRT Rear PTV	53W" 16:9	1080i	Yes	2	2) VGA 15-pin D-sub HD Component	Now	\$6,300.00
Pioneer Elite	PRO-610HD	7" CRT Rear PTV	58W" 16:9	1080i	Yes	2	2) VGA 15-pin D-sub HD Component	Now	\$7,300.00
Pioneer Elite	PRO-710HD	7" CRT Rear PTV	64W" 16:9	1080i	Yes	2	(2) HD Component, Expansion Slot Connection For SH-D07	Now	\$8,300.00
Princeton	AR2.7T	Multi-scan Direct View	27" 4:3	1080i, 720p	Yes	1	(2) RGBHV via VGA	Now	\$1,429.00
Princeton	AR3.2T	Multi-scan Direct View	32" 4:3	1080i, 720p	Yes	1	(2) RGBHV via VGA	Now	\$1,999.00
Princeton	AF3.0HD	Multi-scan Direct View	30W" 16:9	1080i, 720p	Yes	0	RGBHV with BNC or VGA, (2) HD Component	Now	\$4,100.00
Princeton	AS3.6HD	Direct View	36" 4:3	1080i, 720p	Yes	1	RGBHV with BNC or VGA, HD Component	July	\$4,000.00
ProScan	PS52800	Rear Projection	52" 4:3	1080i	Yes	2	RGB 15-pin D-sub, HD Component	Now	\$2,999.00
ProScan	TBA	7" CRT Rear PTV	61" 4:3	1080i	Yes	2	RGB 15-pin D-sub, HD Component	TBA	TBA

Courtesy of TWICE

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1080i and/or 720p-Capable Monitors (15.7kHz-33.75kHz or better) — continued from page 17

Brand	Model	Display Type	Screen Size	HDTV Scan Rate Display Capability	On-Board Line Doubling/Scaling?	Number of NTSC Tuners	Interface for DTV Tuner/Decoder	Available	Price
ProScan	PS32800HR	Multi-scan Direct View	32" 4:3	1080i	Yes	2	RGB 15-pin D-Sub HD Component	Q2-00	\$1,799.00
ProScan	PS36800HR	Multi-scan Direct View	36" 4:3	1080i	Yes	2	RGB 15-pin D-Sub, HD Component	Now	\$2,199.00
ProScan	PS50100W	Plasma Panel	50W" 16:9	720p, 1080i	TBA	TBA	RGB 15-pin D-sub	TBA	TBA
Proton	MM3601VT	Direct View	36" 4:3	1080i	No	0	HD Component VGA 15-pin D-sub	Now	\$3,200.00
Proton	MM2701VT	Direct View	27" 4:3	1080i	No	0	HD Component VGA 15-pin D-sub	Now	\$1,700.00
RCA	MM27100HR	Direct View	27" 4:3	1080i	No	2	15-Pin D-Sub	TBA	\$1,399.00
RCA	MM36100HR	Direct View	36" 4:3	1080i	No	2	15-Pin D-Sub,	Now	\$1,999.00
RCA	MM52100HR	7" CRT Rear PTV	52" 4:3	1080i	Yes	2	15-Pin D-Sub, HD Component	Now	\$2,799.00
RCA	MM61100HR	7" CRT Rear PTV	61" 4:3	1080i	Yes	2	15-Pin D-Sub, HD Component	TBA	\$3,699.00
ReVox	E542	Plasma Panel	42W" (16:9)	480p, 720p (852x480)	Yes	0	Component Video, RGB via VGA	Now	\$18,950.00
Runco	PL-42C	Plasma Panel with controller	42W" 16:9	1080i, 720p, 852x480	Yes	0	HD Component via RCA	Now	\$14,995.00
Runco	PL-50C	Plasma Panel with controller	50W" 16:9	1080i, 720p 1280x768	Yes	0	HD Component	Now	\$25,995.00
Runco Reflection Series	VX-101C	1-chip DLP Front PTV with controller	Variable	720p, 1080i (4:3: 800x600) (16:9: 800x450)	Yes	0	HD RGB	Now	\$9,995.00
Runco Reflection Series	VX-1C	1-chip DLP Front PTV with controller	Variable	720p, 1080i, (4:3: 1024x768) (16:9: 1024x575)	Yes	0	RGBS via VGA 15-pin D-sub	Now	\$16,995.00
Runco Reflection Series	VX-3C	3-chip DLP Front PTV	Variable	720p 1080i (1024x768)	Yes	0	HD Component via BNC	Now	\$64,995.00
Runco	DTV-873	7" CRT Front	Variable	720p 1080i	Scales 540p	0	HD Component via BNC RGB via BNC	Now	\$17,995.00
Runco	DTV-940	7" CRT Front PTV	Variable	720p, 1080i	No	0	HD Component via BNC RGB via BNC	Now	\$15,995.00
Runco	DTV-947	7" CRT Front PTV	Variable	720p, 1080i	Scaler includes controller	0	HD Component via BNC RGB via BNC	Now	\$18,995.00
Runco	DTV-992	8" CRT Front PTV	Variable	720p, 1080i	No	0	HD Component via BNC RGB via BNC	Now	\$27,995.00
Runco	DTV-992 Ultra	8" CRT Front PTV	Variable	720p, 1080i, 1080p	No	0	HD Component via BNC RGB via BNC	Now	\$32,995.00
Runco	DTV-1101	9" CRT Front PTV	Variable	720p, 1080i, 1080p	No	0	HD Component via BNC RGB via BNC	Summer	\$39,995.00
Runco	DTV-5801	7" CRT Rear PTV	58W" 16:9	1080i	Yes	2	HD Component via RCA connectors	Now	\$9,995.00
Runco	DTV-5801c	7" CRT Rear PTV	58W" 16:9	1080i	Yes includes controller	2	HD Component via RCA connectors	Now	\$11,995.00
Samsung Tantus	HCJ552W	7" CRT Rear PTV	55W" 16:9	1080i	Yes	2	HD Component	Now	\$4,999.00
Samsung Tantus	HCJ652W	7" CRT Rear PTV	65W" 16:9	1080i	Yes	2	HD Component	Now	\$6,499.00
Samsung Tantus	PCJ534RF	7" CRT Rear PTV	53" 4:3	1080i	Yes	2	HD Component	Now	\$2,999.00
Samsung Tantus	PCJ614RF	7" CRT Rear PTV	61" 4:3	1080i	Yes	2	HD Component	Now	\$3,499.0
Samsung Tantus	HLK-436W	Ferro LCD Rear PTV	43W" 16:9	All Formats>720p	Yes	2	HD Component, VGA 15-pin D-sub	September	\$5,999.99
Samsung Tantus	HLK-506W	Ferro LCD Rear PTV	50W" 16:9	All Formats>720p	Yes	2	HD Component, VGA 15-pin D-sub	September	\$7,000.00
Seleco	HDPLUS	7" CRT Front PTV	Variable	1080i	No	0	RGBS via BNC, 15-pin D-sub	Now	\$10,995.00
Seleco	SDG-700L6	7" CRT Front PTV	Variable	720p 1080i	No	0	RGBS via BNC, 15-pin D-sub	Now	\$11,595.00
Seleco	SVD-800HD	7" CRT Front PTV	Variable	720p 1080i	Yes (DVDO i-scan)	0	RGBS via BNC, HD Component, 15-pin D-sub	Now	\$15,995.00
Seleco	SVD-800 Millennium	7" CRT Front PTV Hand selected	Variable	720p 1080i	Yes	0	RGBS via BNC, HD Component, 15-pin D-sub	Now	\$19,995.00
Seleco	SDG-900	8" CRT Front PTV	Variable	720p 1080i	No	0	RGBS via BNC, HD Component, 15-pin D-sub	Now	\$19,895.00

Courtesy of TWICE

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1080i and/or 720p-Capable Monitors (15.7kHz-33.75kHz or better) — continued from page 20

Brand	Model	Display Type	Screen Size	HDTV Scan Rate Display Capability	On-Board Line Doubling/Scaling?	Number of NTSC Tuners	Interface for DTV Tuner/Decoder	Available	Price
Seleco	SVD-450	7" CRT Front PTV	Variable	720p 1080i	Scaler	0	RGBS via BNC, HD Component, 15-pin D sub	Now	\$8,995.00
Seleco	SDV-100 Premiere	1-chip DLP Rear PTV	Variable	(800x600)	Scaler	0	RGBS via BNC, HD Component, 15-pin D-sub	Now	\$7,995.00
SharpVision	64LHP4000	7" CRT Rear PTV	64W" (16:9)	1080i	Yes	2	HD Component Video	Now	\$6,995.00
SharpVision	XV-DW100U	LCD Front PTV	Variable	1080i, 720p (1,024x768)	Yes	0	HD Component, RGBHV VGA	Now	\$10,995.00
SharpVision	LC-R60H DU	CG-Silicon LCD Rear PTV	60W" (16:9)	1080i	Yes	0	HD Component RGB H&V sync	Q2-00	\$49,995.00
SharpVision	LC-PD50U	Plasma Panel	50W" (16:9)	720p (1280x768)	Yes	0	HD Component, RGB H&V	Now	\$20,995.00
SharpVision	34N-WF5H	Direct View	34W" (16:9)	1080i	Yes	2	HD Component, RGBHV via VGA	Now	\$4,995.00
SharpVision	50NP4	7" CRT Rear PTV	50" (4:3)	1080i	Yes (to 480p)	2	HD Component	Summer	TBA
SharpVision	60NP4	7" CRT Rear PTV	60" (4:3)	1080i	Yes (to 480p)	2	HD Component	Summer	TBA
Sony	KP-53HS10	7" CRT Rear PTV	53" (4:3)	1080i	Yes (DRC)	2	HD Component	Now	\$3,200.00
Sony	KP-61HS10	7" CRT Rear PTV	61" (4:3)	1080i	Yes (DRC)	2	HD Component	Now	\$3,700.00
Sony	KV-32XBR400	Flat Screen Direct View	32" (4:3)	1080i	Yes (DRC)	2	HD Component	Summer	\$2,200.00
Sony	KV-36XBR400	Flat Screen Direct View	36" (4:3)	1080i	Yes (DRC)	2	HD Component	Summer	\$2,700.00
Sony	KP-53XBR300	7" CRT Rear PTV	53" (4:3)	1080i	Yes (DRC)	2	HD Component	Now	\$4,499.99
Sony	KP-61XBR300	7" CRT Rear PTV	61" (4:3)	1080i	Yes (DRC)	2	HD Component	Now	\$5,499.99
Sony	VPH-G90U	9" CRT Front PTV	Variable	720p 1080i	Yes (DRC)	0	RGBS-BNC, HD Component	Now	\$35,000.00
Sony	VPH-D50HTU	CRT Front PTV	Variable	720p 1080i	Yes (DRC)	0	RGB H&V-sync, HD Component	Now	\$13,990.00
Sony	VPL-VW10HT	LCD Front PTV	Variable	720p (1366x768)	Yes (DRC)	0	RGB H&V-sync HD Component	Now	\$6,990.00
Sony	PFM-510A1WU	Plasma Panel	42W" 16:9	720p, 1080i (1024x1024)	Yes	0	RGBHV, HD Component Video	Now	\$15,999.00
Sony	PFM-510A2WU	Plasma Panel	42W" 16:9	720p, 1080i (1024x1024)	Yes	0	RGBHV, HD Component Video	Now	\$16,999.00
Toshiba	CW34X92	Direct View	34W" 16:9	1080i	Yes (IDSC, horizontal & vertical)	2	2 HD Component Video Inputs	Now	\$4,499.00
Toshiba	CN36X81	Direct View	36" 4:3	1080i	Yes (IDSC, horizontal & vertical)	2	2 sets of HD Component Video	Now	\$2,199.00
Toshiba	TN50X81	7" CRT Rear PTV	50" 4:3	1080i	Yes (IDSC, horizontal & vertical)	2	2 sets of HD Component Video	Now	\$2,999.00
Toshiba	TN55X81	7" CRT Rear PTV	55" 4:3	1080i	Yes (IDSC, horizontal & vertical)	2	2 sets of HD Component Video	Now	\$3,199.00
Toshiba	TN61X81	7" CRT Rear PTV	61" 4:3	1080i	Yes (IDSC, horizontal & vertical)	2	2 sets of HD Component Video	Now	\$3,599.00
Toshiba	TW-56X81	7" CRT Rear PTV	56W" 16:9	1080i	Yes (IDSC, horizontal & vertical)	2	2 sets of HD Component Video	Now	\$4,999.00
Toshiba	TW-65X81	7" CRT Rear PTV	65W" 16:9	1080i	Yes (IDSC, horizontal & vertical)	2	2 sets of HD Component Video	Now	\$6,499.00
Toshiba	TW-40X81	7" CRT Rear PTV	40W" 16:9	1080i	Yes (IDSC, horizontal & vertical)	2	2 sets of HD Component Video	Now	\$2,999.00
Vidikron	Vision One	CRT Front PTV	Variable	720p 1080i 1080p	No	0	RGBS via BNC	Now	\$49,995.00
Vidikron	Vision Two	CRT Front PTV	Variable	720p 1080i 1080p	No	0	RGBS via BNC	Now	\$31,995.00
Vidikron	Vision Three	CRT Front PTV	Variable	720p 1080i 1080p	No	0	RGBS via BNC	Now	\$24,995.00
Vidikron	VPF50HDX	CRT Front PTV	Variable	720p 1080i 1080p	No	0	RGBS BNC	Now	\$19,995.00
Vidikron	Epoch D-600	LCD Front PTV	Variable	720p 1080i (1024x768)	Digital Scaler	0	RGB via VGA 15-pin D-Sub	Now	\$9,495.00
Vidikron	Epoch D-2200	LCD Front PTV	Variable	720p 1080i 1080p	Digital Scaler	0	(2) RGB via VGA HD Component	Now	\$13,495.00
Vidikron	Kronos One	7" CRT Front PTV	Variable	720p 1080i	No	0	RGBS via BNC	Now	\$10,995.00
Zenith	IQC50H94W	7" CRT Rear PTV	50 (4:3)	1080i	Yes	2	HD Component	Q3-00	TBA
Zenith	IQC50H95W	7" CRT Rear PTV	50 (4:3)	1080i	Yes	2	HD Component	Q3-00	TBA
Zenith	IQC60H94W	7" CRT Rear PTV	60 (4:3)	1080i	Yes	2	HD Component	Q3-00	TBA
Zenith	IQC60H95W	7" CRT Rear PTV	60 (4:3)	1080i	Yes	2	HD Component	Q3-00	TBA
Zenith	IQD61W20	7" CRT Rear PTV	61 (16:9)	1080i	Yes (Enhanced)	2	HD Component, RGB H/V via VGA 15-pin D-sub	Q4-00	TBA
Zenith	Pro 900X	7" CRT Front PTV	Variable	720p 1080i	No	0	RGB H&V-sync	Now	\$12,600.00
Zenith	Pro 1200X	8" CRT Front PTV	Variable	720p 1080i	Optional add-in line-quadrupler	0	RGB H&V-sync	Mid Year	\$24,995.00

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TECHNOLOGY UPDATE

Solutions for Linking Cable and DTV

Approximately 70 percent of American homes currently receive their TV programming through a cable system. For this reason, it is critical to ensure that the full benefits of digital and high-definition television reach cable subscribers. This requires technical solutions that link digital television sets to cable set-top boxes and government policies that enforce the cable industry's obligation to carry high-definition programming.

Cable Compatibility Agreements Reached

After numerous productive discussions between the Consumer Electronics Association (CEA) and the National Cable Television Association (NCTA) over the past year, the two groups announced in February that they had reached voluntary agreements that allow digital television (DTV) sets and digital cable systems to work together. In May, the industries announced further agreement on labeling of DTV products with regard to cable compatibility. These agreements are important milestones in the U.S. transition to digital television because they will allow consumers to receive DTV programming and services over cable systems.

Specifically, the agreements spell out the technical requirements that permit the direct connection of digi-

tal television receivers to cable television systems, and specify the signal levels and quality as well as video formats. They also provide for the carriage of Program and System Information Protocol (PSIP) data on cable systems to support on-screen guide functions in digital receivers. Subject to certain conditions, PSIP data enables features such as on-screen program guides, virtual channel tables, program name and description (for a minimum 12-hour period) and content advisory information.

The May agreement between the cable and consumer electronics industry acknowledged that every digital TV set will not need to include

a 1394/5C connector allowing reception of the full range of cable interactive services. However, all sets will be packaged with consumer information describing the features and functions of television sets with and without the 1394/5C connector. The descriptive information will appear in consumer electronics product manuals and brochures.

Sets labeled "Digital TV-Cable Connect" - those without the 1394/5C connector - will be capable of receiving analog basic, digital basic and digital premium cable pro-

gramming from any cable system that offers digital service. "Digital TV-Cable Interactive" sets - those with the 1394/5C connector - will be able to receive those services and other programming, including impulse pay-per-view, video-on-demand, enhanced program guides and data enhanced television services with a digital set top box. CEA

and NCTA have agreed to continue discussions and expect to reach an agreement on the labeling of digital set-top boxes that will work with the "Digital TV-Cable Interactive" DTV sets.

CEA estimates that the first digital TV

receivers bearing the new labels will reach market by the fourth quarter of 2001.

Copy Protection Update

The transition to digital television requires new solutions for protecting content because the interface between two or more consumer electronics products that carry digital video and audio is a potential target for unauthorized copying. The consumer electronics, cable, motion picture, recording and information technology industries are working to

CEA has developed four technical solutions for linking cable and other set-top boxes to television receivers:

- *1394 interface,*
- *RF remodulator interface,*
- *component video interface,*
- and
- *the National Renewable Security Standard (NRSS) interface.*

These four interfaces provide consumers and manufacturers a wide range of choices for connecting their digital TV sets to cable systems.

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Fully Integrated 1080i Sets

(Digital Decoder included)

Brand	Model	Display Type	Screen Size	Aspect Ratio	DTV Native Scan Format	Scan Conversion	Line Doubling For NTSC?	Built-In AC-3 Decoder?	IEEE 1394 Digital Interface	Available	Suggested Retail Price
Daewoo	DSP-3060N	Direct View	30W	16:9	1080i	All Formats>1080i	Yes	Yes	No	TBA	\$3,000.00
Hitachi	61HDX98B*	7" CRT Rear Projection	61W	16:9	1080i	NTSC>540p 480i>540p 480p>540p 720p>1080i 1080i>1080i	Yes	Yes	No	Now	\$5999 includes dish
Hitachi	61HDX01W*	7" CRT Rear Projection	61W	16:9	1080i	NTSC>540p 480i>540p 480p>540p 720p>1080i 1080i>1080i	Yes	Yes	No, includes HD component input	September	\$5,999 includes dish
Konka	HD3098U	Direct View	30W"	16:9	1080i	All Formats>1080i	Yes	Yes	No	TBA	\$3,499.00
Konka	HD3498U	Direct View	34"	16:9	1080i	All Formats>1080i	Yes	Yes	No	TBA	TBA
Marantz	HD6400W	9" CRT Rear Projection	64W"	16:9	1080i	NTSC>525p 480i>1080i 480p>1080i 720p>1080i 1080i>1080i	Yes	Yes	No	September	\$11,999.00
Philips	64PH9905	9" CRT Rear Projection	64W"	16:9	1080i	NTSC>525p 480i>1080i 480p>1080i 720p>1080i 1080i>1080i	Yes	Yes	No	Now	\$9,990.00
ProScan	PS65000	7" CRT Rear Projection	65W	16:9	1080i	NTSC>540p 480i>540p 480p>540p 720p>1080i 1080i>1080i	Yes	Yes	No	Now	\$5,499.00
ProScan	PS34000 Performax	Direct View	34W"	16:9	1080i	NTSC>540p 480i>540p 480p>540p 720p>1080i 1080i>1080i	Yes	Yes	No	TBA	\$3,499.00
ProScan	PS38000 Performax	Direct View	38W"	16:9	1080i	NTSC>540p 480i>540p 480p>540p 720p>1080i 1080i>1080i	Yes	Yes	No	August	\$3,999.00
RCA	F38310	Direct View Performax	38W"	16:9	1080i	NTSC>540p 480i>540p 480p>540p 720p>1080i 1080i>1080i	Yes	Yes	No	August	\$3,799.00
RCA	P61310*	7" CRT Rear Projection	61W"	16:9	1080i	NTSC>540p 480i>540p 480p>540p 720p>1080i 1080i>1080i	Yes	Yes	No	Now	\$4,999.00
Samsung (Tantus Digital)	HCJ555W	7" CRT Rear Projection	55W"	16:9	1080i	All Formats>1080i	Yes Includes HD Component	Yes	No	Now	\$6,999.00
Samsung (Tantus Digital)	HCJ655W	9" CRT Rear Projection	65W"	16:9	1080i	All Formats>1080i	Yes Includes HD Component	Yes	No	Now	\$9,999.95
SharpVision	64LHP5000	7" CRT Rear Projection	64W"	16:9	1080i	NTSC>480p 480i>480p 720p>1080i 1080i>1080i	Yes	Yes	No	Now	\$8,995.00
Sanyo	TBA	Direct View	30W"	16:9	1080i	NTSC>480i/p 480i>480i/p 480p>480p 720p>1080i 1080i>1080i	Yes	Yes	No	TBA	\$3,499.00
Sony	KWP-65HD1	CRT Rear Projection	65W"	16:9	1080i	NTSC>960i 480i>960i 480p>480p 720p>1080i 1080i>1080i	Yes (DRC)	No on-board ProLogic	No	Now	\$9,999.00
Sony	KW-34HD1	Direct View	34W"	16:9	1080i	NTSC>960i 480i>960i 480p>480p 720p>1080i 1080i>1080i	Yes	Yes	No	Now	\$7,499.00
Toshiba	DW65X91*	7" CRT Rear Projection	65W"	16:9	1080i	NTSC>540p 480i>540p 480p>540p 720p>1080i 1080i>1080i	Yes Includes HD Component inputs	Yes	No	Now	\$7,999 includes dish
Zenith/Inteq	IQB64W10W	9" CRT Rear Projection	64W"	16:9	1080i	All Formats>1080i	Yes Includes HD Component inputs	Yes	No	Now	\$9,999.00
Zenith/Inteq	IQD56W10G	7" CRT Rear Projection	56W"	16:9	1080i	All Formats>1080i	Yes Includes HD-compatible video, RGB via VGA inputs	Yes	No	Now	\$7,999.00
*Also receives DirecTv standard and HD satellite programming.											

Courtesy of TWICE

480p CAPABLE MONITORS

(31.5kHz or higher scanning frequency) External DTV-Decoder Required

Brand	Model	Display Type	Screen Size	Aspect Ratio	DTV Native Scan Format	On-Board Line Doubling/Scaling?	Number of NTSC Tuners	Interface for DTV Tuner/Decoder	Available	Price
Fujitsu	PDS-4203	Plasma Panel	42W"	(16:9)	480p	Yes	0	RGB-H&V BNC, Component BNC, RGB 15-pin D SUB	Now	\$6,995.00
Fujitsu	PDS-4212	Plasma Panel	42W"	(16:9)	480p	Yes	0	RGB-H&V BNC, Component BNC, RGB 15-pin D SUB	Now	\$9,995.00
Hitachi	52LDX99B	LCD Rear PTV	52"	(4:3)	Non-standard 600p (4:3) 450p (16:9)	Yes Flex Converter	2	HD Component, VGA 15-pin D-sub	June	\$4,999.00
InFocus	LS-700	LCD Front PTV	Variable	Variable	480p (XGA)	Yes Processing by Faroudja	0	VGA	Now	\$12,995
Loewe	Planus Tabletop	Direct View	30W"	(16:9)	480p	Yes	2	RGB via VGA	Now	\$3,800.00
Loewe	Planus Console	Direct View	30W"	(16:9)	480p	Yes	2	RGB via VGA	Now	\$4,400.00
Loewe	Arcada Tabletop	Direct View	32"	(4:3)	480p	Yes	2	RGB via VGA	Now	\$2,800.00
Loewe	Calida Tabletop	Direct View	32"	(4:3)	480p	Yes	2	RGB via VGA	Now	\$2,600.00
Loewe	Art Console	Direct View	36"	(4:3)	480p	Yes	2	RGB via VGA	Now	\$4,500.00
Marantz	PD-4290D	Plasma Panel	42W"	(16:9)	1080i, 720p (853x480)	Yes	0	RGB via VGA, HD Component	Now	\$14,999.0
NetTV	DTV42WP	Plasma Panel	42W"	(16:9)	480p	Yes	0	RGB via VGA	Now	\$8,999.00
Panasonic	PT-61DX80	7" Rear PTV	61"	(4:3)	480p	Yes	2	HD Component Video	Now	\$3,199.00
Panasonic	PT-51DX80	7" Rear PTV	51"	(4:3)	480p	Yes	2	HD Component Video	Now	\$2,699.00
Panasonic	CT-32XF56	Direct View	32"	(4:3)	480p	Yes	2	HD Component Video	Now	\$1,799.00
Panasonic	PT-42PD2-P	Plasma Panel	42W"	(16:9)	480p	Yes	0	HD Component, VGA 15-pin D-sub	September	TBA
Philips	42FD9932	Plasma Panel	42W"	(16:9)	480p	Yes	0	RGB via 15-pin D-sub, HD Component	Now	\$9,990.00
ProScan	PSP4200	Plasma Panel	42W"	(16:9)	480p	Yes	0	RGB via VGA	Now	\$12,999.00
Samsung	SLK-407W	TFT LCD Rear PTV	40W"	(16:9)	480p native	Yes	2	HD Component Video VGA	June	\$3,999.00
Samsung	TSK2792F	Flat Face Direct View "PureFlat"	27"	(4:3)	480p native	Yes	2	HD Component Video	August	\$1,199.00
Samsung	TSK3293F	Flat Face Direct View "PureFlat"	32"	(4:3)	480p native	Yes	2	HD Component Video	August	\$1,699.00
Samsung	TSK3092F	Flat Face Direct View "PureFlat"	30W"	(16:9)	480p	Yes	2	HD Component Video	August	\$2,199.00
SharpVision	XVZW99	LCD Front PTV	Variable	Variable	480p	Yes	0	RGB H&V, HD Component	Now	\$7,999.00
Sony	PFM-500A2WU	Plasma Panel	42W"	(16:9) (852x480)	480p	Yes	0	RGBHV HD Component	Now	\$9,999.0

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develop copy protection solutions that will defeat copy piracy while promoting compatibility between DTVs, digital VCRs and set-top boxes. The copy protection issue is the subject of a current FCC rule-making proceeding.

To date, the industry has developed a number of legitimate and effective approaches to protect against unauthorized copying of digital broadcast programming and pre-

recorded content on consumer electronics devices. One solution, referred to as the "5 C" or "5 Company" proposal, was put forth by Hitachi, Intel, Matsushita, Sony, and Toshiba. Another, called the "Extended Conditional Access (XCA)", was put forth by Thomson and Zenith. Manufacturers and content providers may have additional choices for copy protection as manufacturers continue to develop innovative technologies to provide protection across an array of

digital interfaces.

It is important to remember that the copy protection issue has not prevented content providers from offering a variety of digital and high-definition programming. Broadcasters have already aired feature films, major sporting events and documentaries on network television and major satellite providers are distributing a wide array of digital programming from HBO and other programming sources. ■

BROADCASTERS SEE DATA COMING TO DTV

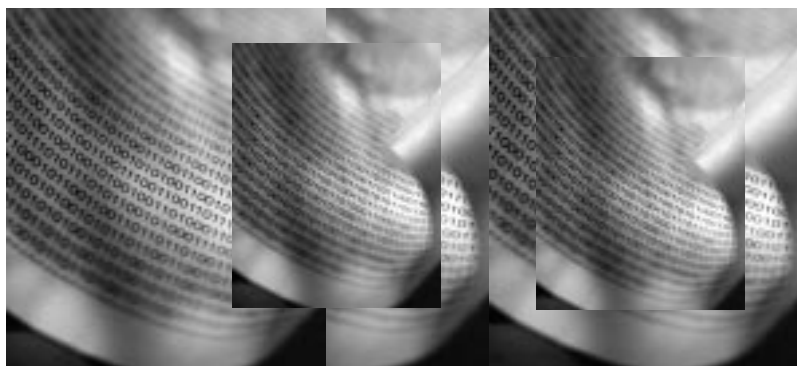
By Glen Dickson

Beyond delivering the straightforward promise of movie-quality pictures and sound in HDTV programs, broadcasters are exploring additional future opportunities for their new digital spectrum.

As digital compression technology improves rapidly, stations are moving to include interactive entertainment and communications services in addition to HDTV content over their 19.4 Mb/s DTV pipe.

"The digital standard offers tremendous flexibility," says Dick Wiley, former Federal Communications Commission (FCC) chairman and former chairman of the Advisory Committee on Advanced Television Service (ACATS). "While some may want to limit it, there's no reason to. By no means should it be an either-or choice between the various services available. The consumer should have access to all of the services, including high-definition, standard-definition, and a variety of interactive offerings for shadowing the union of the TV set and the personal computer."

Most broadcast engineers feel they can transmit full-quality HDTV pictures, even the bandwidth-intensive 1080-line interlace format, in 12-14 Mb/s. That frees up an additional 5-7 Mb/s for other services, such as standard-definition programming or data services.



"We need to encourage people to think of the DTV channel as a large pipeline to the consumer," says Andy Butler, Senior Director of Engineering for PBS, who says PBS's engineering lab is "getting very good HD" at 7 to 8 Mb/s in its tests of next-generation encoders.

"We're going to get better service-per-bit efficiency on a very accelerated basis," says Butler. "You can't afford to limit your dreaming on what you can get today."

As the demand for Internet access and multimedia content grows at an explosive rate, stations increasingly see DTV as a way to compete with cable operators, phone companies and direct broadcast satellite (DBS) in the broadband race.

In fact, many broadcasters feel they are uniquely positioned to deliver some of the most popular content on the Web, particularly streaming media events that gobble up a lot of bandwidth and need to reach a mass audience. While Internet distribution companies like iBeam and Akamai look to use satellites to bypass Internet bottlenecks until the last mile to the home, broadcasters point out that

they can deliver data all the way to the home without ever relying on a landline.

"The Internet backbone is having trouble scaling to broadband use," notes Rick Ducey, president of datacasting firm SpectraRep. He cites the recent Victoria's Secret online fashion show, when five million users flooded the site, as a good example of the Internet's limitations.

"The telephone system is premised on the idea of 1 in 5 people using it, and cable 1 in 10," says Ducey. "Terrestrial networks are not 'multicast-enabled'—they can't broadcast to many users."

Unlike cable modems, the DTV bandwidth dedicated to an individual consumer won't decrease when his neighbors decide to share the broadband experience. And unlike DSL, whose geographic reach is haphazard and limited by distance from upgraded switching centers, every home in a broadcast market that can receive a DTV signal will be able to enjoy high-speed wireless data.

There are a few caveats, of course. The very nature of broad-

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DTV Technology

Digital TV Set Top Decoders

Brand	Model	DTV Formats Received	Scan Conversion (Input>Output)	HD Interface for DTV Monitor	Includes NTSC Receiver	Includes NTSC Line Doubler/Scaler	Built in Dolby Digital Decoder	IEEE 1394 Digital Interface	Available	Suggested Retail Price
EchoStar	Adapter module for Dish 5000**	Core 18 Table 3 Formats	Passes Undecoded HD Signal In Format Received	Connects via RF (RG6) To Separate DTV Decoder	No	No	No	No	Now	\$299.00
EchoStar	6000**	Core 18 Table 3 Formats	All Formats>480i All Formats>1080i All Formats>720p	HD Component, RGB H&V via BNC, RGB via 15-pin D-Sub	Has Upgradable Expansion Slot For Optional Terrestrial NTSC/ATSC Tuners	No	Yes (2 Channel)	No	Now	\$499.00; optional ATSC tuner module \$99
EchoStar	6100**	Core 18 Table 3 Formats	All Formats>480i All Formats>1080i All Formats>720p	HD Component, RGB H&V via BNC, RGB via 15-pin D-Sub	Receives both NTSC and ATSC Terrestrial Broadcasts	No	Yes (2 Channel)	No	TBA	TBA
General Instruments	HDD200*** Adapter module for 4DTV digital C-Band Decoder	C-Band only ; No ATSC tuner included	Selectable: All Formats>480p All Formats>720p All Formats>1080i	HD Component, RGBHV	No	Yes, multiplies externally tuned NTSC signals	No	No	Now	\$1,599.00
Hughes Network Systems	HSYE-4686 Platinum HD*	Core 18 Table 3 Formats	All Formats>480i All Formats>1080i HD Component	HD Component	Yes	No	Yes	No	Late July	TBA
Integra	IT815ST Formerly Unity Motion HDR-1000A	Core 18 Table 3 Formats	HD Formats>1080i SD Formats>480p	HD Component RGBHV D-sub	No	No	Yes	No	Now	\$795.00
Konka	HD-0001	Core 18 Table 3 Formats	All Formats>1080i	RGB (High Density 15-pin D-sub)	No	No	Yes	No	Nov-2000	\$999.95
Loewe	TBA	Core 18 Table 3 Formats	Switchable: All Formats>480i All Formats>480p All Formats>1080i	RGB via VGA, HD Component Video	Yes	No	Yes (2-channel)	No	TBA	TBA
Mitsubishi	HD-1080	Core 18 Table 3 Formats	480i>960i 480p>960i 720p>1080i 1080i>1080i	Proprietary RGB H&V-Sync	No	Yes	Yes (2-channel)	No	Now	\$1,000.00
Mitsubishi	SR-HD500*	Core 18 Table 3 Formats	All Formats>480i All Formats>1080i	HD Component Video RGB H&V-Sync	Yes	No	Yes (2-channel)	Yes	August	\$1,199.00; includes dish
Mitsubishi	SR-HD400*	Core 18 Table 3 Formats	All Formats>480p All Formats>1080i	HD Component, RGB, H&V-Sync	Yes	No	Yes (2-channel)	No	August	\$1,099.00
Panasonic	TU-HDS20*	Core 18 Table 3 Formats	All Formats>Any Output Selected	Switchable: HD Component or RGB H-V (RCA jacks)	Yes	Yes	No	No	Now	\$1,099.95
Panasonic	TU-DST51	Core 18 Table 3 Formats	Switchable: All Formats>NTSC, All Formats In Native Form, 720p>480p, 1080i>480p, 480i>480p	HD Component Video	Yes	No	No	Yes	Now	\$1,099.95
Panasonic	TU-DA2420	Dish for the TU-HDS20	NA	NA	NA	NA	NA	NA	Now	\$199.00
Pioneer	SH-D505	Core 18 Table 3 Formats	Switchable: All Formats>480p All Formats>720p All Formats>1080i	HD Component Video, RGB H&V-Sync	No	No	No	Upgradable Expansion Port	Now	\$2,500.00
Pioneer	SH-D09	Core 18 Table 3 Formats	Switchable: All Formats>480p All Formats>720p All Formats>1080i	Expansion Slot Connection for Pro-700HD	No	No	Yes	Upgradable Expansion Port	Now	\$2,500.00
ProScan	PSHD105*	Core 18 Table 3 Formats	Switchable: All Formats>480i All Formats>540p, 720p>1080i, 1080i>1080i	RGB via VGA	Yes	Yes	Yes	No	TBA	\$649.00

Courtesy of TWICE

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DATA COMING TO DTV

casting is that it is a one-way medium: from a single transmitter to many receivers (also known as point-to-multipoint). Broadcasters will have to partner with other telecommunications providers, such as a consumer's existing Internet Service Provider (ISP), if they require a backchannel for their new datacasting services. And broadcasters don't currently have the billing infrastructure or customer service personnel in place to handle a new subscription business. Speaking of subscription revenues, there is no free lunch with DTV data—the FCC has already said that broadcasters will have to give the government 5% of any new revenues generated by subscription services on their DTV spectrum.

Nonetheless, a good number of broadcasters and technology firms believe the pros of datacasting far outweigh the cons and are vigorously pursuing these new services, which will primarily be aimed at personal computers equipped with DTV receiver cards and small indoor antennas.

Here are the major players:

Geocast Network Systems of Menlo Park, Calif. is a start-up company founded by veterans of both the broadcasting and computer industries. Its aim is to create a national multimedia service, distributed by satellite to local markets and then retransmitted by local DTV stations, that will encompass national content, such as financial information, movie trailers and music videos, as well as local news and information from broadcast sta-

tions. The company will place equipment and personnel at the local station to receive the national service, integrate local content, and turn it around for local broadcast.

Geocast hopes to generate advertising revenue and will also pursue transactional revenues by relying on a consumer's existing ISP as a backchannel for e-commerce. For example, Geocast might download an electronic copy of the J. Crew catalog to a consumer in return for a cut of any orders that consumer may place with J. Crew. The company also thinks it can download software programs for a fee; at the NAB 2000 convention in Las Vegas, Geocast demonstrated the complete download of Microsoft Money, a 40 MB personal finance program, in just 15 seconds.

For a company with an unconventional idea, Geocast has amassed a solid financial base in addition to a wealth of strategic partnerships. It has raised over \$200 million in funding, mostly from venture capital firms, and is already constructing a satellite uplink facility in California to beam its service across the country. The company has struck content deals with three major broadcast station groups, Hearst-Argyle, A.H. Belo and Allbritton Communications, as well as cable programming powerhouse Liberty Digital and game developer Electronic Arts. All of Geocast's broadcast and content partners have invested between \$1 million and \$10 million in the firm. More significantly, the station groups have pledged part of their DTV spectrum—a daily average of 6 Mb/s, with a minimum contribution of 2.5

Mb/s—to the venture.

Geocast's receiver model is unique in that it doesn't plan to rely on standard PCs and plug-in DTV receiver cards. Instead, it has designed a PC peripheral device that will include multiple 8-VSB tuners, so as to receive several stations in a market simultaneously, and a large amount of disk storage to cache multimedia content. The \$299 box will be manufactured by Thomson Consumer Electronics, which has also invested in Geocast. Thomson showed a prototype of the Geocast receiver at CES this year, and Geocast plans to launch its service in early 2001.

Another aggressive datacasting player, albeit with a completely different business plan, is Los Angeles-based iBlast. The company was formed in early March by 12 major station groups: Tribune Broadcasting, Gannett Broadcasting, Cox Broadcasting, Post-Newsweek Stations, E.W. Scripps, Meredith Broadcasting, Media General Broadcasting, Lee Enterprises, The New York Times Co., McGraw-Hill Broadcasting, Smith Broadcasting and Northwest Broadcasting. Each partner station—143 in all—has committed a daily average of 7 Mb/s of its DTV bandwidth. iBlast's stations will reach 102 markets covering 80% of U.S. homes, and at press time the company said it was negotiating with several more station groups for deals that would significantly increase its national footprint.

iBlast's business plan is simple: it thinks it can generate significant revenues by providing a transport

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L.A.'S DTV DESTINATION

sumer interest in the system and drives people into the stores to have a look, and once many of them get a taste of HDTV, they want it right away.

Ken Crane's is dedicating a large percentage of its floor space to digital TV and much of our advertising now talks about the quality and excitement of the HDTV experience, including on-air testimonials from our customers.

I believe demand for HDTV upgradeable monitors, set-top digital decoder boxes and even some fully integrated HDTV sets will continue to produce unseasonably strong sales volume throughout the year and into next year. Rarely have we seen so much momentum for a product category this early in the game. Our only limitation will be getting enough supply to meet our needs, as manufacturers continue to gauge their production to the continued uncertainty over some unsettled standards issues.

I am happy to say, the established standards in the ATSC system have worked well in our market. We rarely have a problem with signal reception when a customer's system is properly installed, and we have not seen any buyer's remorse. The satisfaction rate is second to none. ■

Tom Campbell is corporate director for Ken Crane's Home Entertainment City Inc. and senior technology director for the 2000 Presidential Debates.

Digital TV Set Top Decoders—continued from page 28

Brand	Model	DTV Formats Received	Scan Conversion (Input-Output)	HD Interface for DTV Monitor	Includes NTSC Receiver	Includes NTSC Line Doubler/Scaler	Built in Dolby Digital Decoder	IEEE 1394 Digital Interface	Available	Suggested Retail Price
Proton	TBA	Core 18 Table 3 Formats	Switchable: All Formats>480i All Formats In Native Form, 720p>480p, 1080i>480p, 480i>480p	RGB via VGA, Component (YUV)	No	No	Yes	No	TBA	TBA
RCA	DTC100*	Core 18 Table 3 Formats	Switchable: All Formats>480i, All Formats>540p, 720p>1080i, 1080i>1080i	RGB via VGA	Yes	Yes	Yes	No	Now	\$649.00 without dish
Samsung	SIR-T100	Core 18 Table 3 Formats	Switchable: All Formats>480p All Formats>1080i	HD Component Video	No	Yes	Yes	No	Now	\$1,999.00
Samsung	SIR-T150	Core 18 Table 3 Formats	Switchable: All Formats>480p All Formats>720p All Formats>1080i All Formats>NTSC Line doubles NTSC	HD Component Video RGB via VGA	No	No	No	No	Sept. 2000	\$699.00
Samsung	SIR-TS200*	Core 18 Table 3 Formats	Switchable: All Formats>480p All Formats>720p All Formats>1080i All Formats>NTSC Line doubles NTSC	HD Component Video RGB via RCA	No	No	No	No	Spring 2001	TBAI
SharpVision	TU-DTV1000	Core 18 Table 3 Formats	Switchable: All Formats>480i All Formats>480p All Formats>1080i	HD Component Video RGB H&V-Sync	No	No	Yes	Upgradable Expansion Port	Now	\$1,995.00
Sony	SAT-HD100*	Core 18 Table 3 Formats	Switchable: All Formats>480i All Formats>1080i	HD Component	Yes	No	No "DD compatible"	No	August	\$699.00
Toshiba	DST-3000*	Core 18 Table 3 Formats	Switchable: All Formats>480i All Formats>1080i	HD Component Video	Yes	No	Yes	No	August	\$999 w/o dish
Zenith	IQDTV-1080*	Core 18 Table 3 Formats	All Formats>1080i, 720p, 480p, 480i All Formats>NTSC	RGB via VGA HD Component	Yes	Yes (enhanced)	Yes	No	Q4-00	TBA

Courtesy of TWICE

DTV Retail

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BEST BUY TAKES DTV LEAD

visions within the consideration band of affordable television. Today, our customers can purchase a true high-quality digital-compatible television set for between \$2,000-\$3,400.

Clearly, digital television is the wave of the future, and personally, I would like to see more of a focus on HDTV programming from the broadcasters to help us keep the momentum going.

Unfortunately, access to high definition or even standard definition digital television signals is not consistent in all markets across the country. At our Minneapolis headquarters, we still only have one station (PBS) at full power and one (ABC) station at test power broadcasting digitally. We are a year behind schedule for all network stations. Obviously, as more content

becomes available, more people will be willing to buy digital television products.

Right now, people are buying digital-capable televisions for what they can view today. Many of these new TVs incorporate some form of picture enhancement circuitry that boosts the quality of analog source material to take advantage of the higher scan rates of these new television displays. The result is a picture that looks close in quality to a digital broadcasts, but using content that originating from traditional NTSC-based components, such as DVD players, VCRs, home satellite and cable sources.

Consumers also purchase digital television products today to avoid obsolescence. They know that digital TV broadcasting is coming, and they want to be ready for it.

Looking ahead, this Holiday season should generate greater growth of digital and HDTV-compatible products. I have been told that supply shortages should not be an issue, although we know from past experience that today's word on digital TV does not necessarily result in tomorrow's delivery.

I look for Best Buy's sales of projection televisions to be evenly split between digital and analog models in 2001.

I also expect to see increased sales in set-top digital TV tuner/decoders that will enable these digital-capable models to receive off-air standard- and high-definition programming. We should also begin to see more interest in fully integrated HDTV sets that have built-in, off-air digital television tuners.

So far we have sold a relatively

small number of fully integrated models, but sales of set-top digital TV decoder boxes have started to grow. Currently, we carry RCA's DTC100 off-air digital TV decoder box with an integrated DirecTV standard and high-definition TV satellite signal decoder. I expect sales for this type of product to increase as more television stations begin broadcasting high definition and multicast digital TV channels.

Unfortunately, we will not be receiving new fully integrated HDTV sets from Sony this year. But as an industry, I think we might have anticipated that, and should be able to produce enough products to meet the demand. The company's decision to delay shipment is not surprising, when you consider we have no solid standards in place for digital copy protection to go with a digital IEEE-1394 connection or a protocol language for networked system components. We also face an uncertain status over the current digital broadcasting standard as the Federal Communications Commission fields criticisms over the modulation scheme it had previously approved.

Personally, I don't know what the industry will do if that standard is revised. I assume some of the fully integrated digital sets and set-top boxes we've sold already will be able to receive another modulation standard. If not, there will be significant reworking to do, at the risk of alienating every customer who bought one of these sets over last two years. It could also cause other consumers significant apprehension about buying a digital television of any form in the future. ■





Stations Broadcasting Digital Signal

Station	City	Date On	Owner	Signal Type	Format	Time of Day	Cable Activity
WCBS (CBS)	New York	Nov. 1 98	CBS	High definition	1080i	24 hours	Digital signal carried on Time-Warner system
WNYW (Fox)	New York	April, 99	Fox	Standard definition	480p	24 hours	NA
KTLA (WB)	Los Angeles	Oct. 98	Tribune	High definition	480p	12 hours	NA
KCBS (CBS)	Los Angeles	Nov. 98	CBS	High definition	1080i	24 hours	Digital signal carried on Time-Warner system
KNBC (NBC)	Los Angeles	Oct. 98	NBC	High definition	1080i	24 hours	NA
KABC (ABC)	Los Angeles	Nov. 98	ABC	High definition	720p	24 hours	NA
KABC (UPN)	Los Angeles	Dec. 98	Chris Craft	Standard definition	480p	24 hours	NA
KTTV (Fox)	Los Angeles	May, 99	Fox	Standard definition	480p	24 hours	NA
KCAL (Indp.)	Los Angeles	Sept. 99	Young Broadcast.	Standard definition	480i	24 hours	NA
KCET (PBS)	Los Angeles	May 2000	Public	High Definition	1080i	24hours	NA
WFLD (Fox)	Chicago	May, 99	Fox	High definition	720p	24 hours	NA
WCPC (Pax)	Chicago	Feb. 2000	Paxson	Standard definition	480i	24 hours	NA
WPVI (ABC)	Philadelphia	Nov. 98	ABC	High definition	720p	24 hours	NA
KYW (CBS)	Philadelphia	Nov. 98	CBS	High definition	1080i	24 hours	Digital signal carried on Time-Warner system
WTFX (FOX)	Philadelphia	Oct. 98	Fox	High definition	720p	24 hours	NA
WCAU (NBC)	Philadelphia	Nov. 98	NBC	High definition	1080i	24 hours	NA
WHYY (PBS)	Philadelphia	Nay 2000	Public	High definition	NA	NA	NA
KGO (ABC)	San Francisco	Nov. 98	ABC	High definition	720p	24 hours	NA
KPIX (CBS)	San Francisco	Nov. 98	CBS	High definition	1080i	24 hours	NA
KTVU (Fox)	San Francisco	Nov. 98	Cox	High definition	720p	24 hours	NA
KRON (NBC)	San Francisco	Nov. 98	Chronicle	High definition	1080i	24 hours	NA
KBHK (UPN)	San Francisco	June, 99	Chris Craft	Standard definition	480p	24 hours	NA
KBWB (WB)	San Francisco	March 2000	Granite Brcdcast.	Standard definition	NA	NA	NA
KQED (PBS)	San Francisco	May 2000	Public	High definition	1080i	20 hours	NA
KICU (Indep)	San Jose	Dec. 99	Cox	Standard definition	480i	24 hours	NA
WCVB (ABC)	Boston	Oct. 98	Hearst-Argyle	High definition	720p	24 hours	NA
WMUR (ABC)	Boston	Nov. 98	Imes	Standard definition	480i	24 hours	Ongoing technical trial with MediaOne
WHSN (HSN)	Boston	Nov. 98	USA Broadcast.	Standard definition	480p	8 hours	NA
WHDH (NBC)	Boston	April, 99	Sunbeam Tel.	Standard definition	480i	24 hours	NA
WFXT (Fox)	Boston	April, 99	Fox	High definition	720p	24 hours	NA
WBZ (CBS)	Boston	May, 99	CBS	High definition	1080i	24 hours	NA
WFAA (ABC)	Dallas	Nov. 98	Belo	High definition	1080i	24 hours	NA
KDFW (Fox)	Dallas	Nov. 98	Fox	High definition	720p	24 hours	NA
KXAS (NBC)	Dallas	Nov. 98	NBC	High definition	1080i	24 hours	NA
KTVT (CBS)	Fort Worth	May, 99	Gaylord	High definition	1080i	24 hours	NA
WJLA (ABC)	Washington	Nov. 98	Allbritton	Standard definition	480p	24 hours	NA
WUSA (CBS)	Washington	Nov. 98	Gannett	High definition	1090i	24 hours	NA
WRC (NBC)	Washington	Nov. 98	NBC	High definition	1080i	24 hours	NA
WETA (PBS)	Washington	Nov. 98	Public	High definition	1080i	24 hours	NA
WTTG (Fox)	Washington	July, 99	NA	NA	NA	NA	NA
WXYZ (ABC)	Detroit	Oct. 98	Scripps Howard	High definition	720p	17 hours	NA
WJBK (Fox)	Detroit	May, 99	Fox	High definition	720p	24 hours	NA
WDIV (NBC)	Detroit	April, 99	Post Newsweek	High definition	1080i	24 hours	NA
WWJ (CBS)	Detroit	July, 99	CBS	High definition	1080i	24 hours	NA
WSB (ABC)	Atlanta	Nov. 98	Cox	High definition	720p	24 hours	NA
WXIA (NBC)	Atlanta	Nov. 98	Gannett	High definition	1080i	24 hours	NA
WGNX (CBS)	Atlanta	NA	Tribune/Meredith	High definition	1080i	24 hours	NA
WAGA (Fox)	Atlanta	April, 99	Fox	High definition	720p	24 hours	NA
KHOU (CBS)	Houston	Nov. 98	Belo	High definition	1080i	24 hours	NA
KRIV (FOX)	Houston	Aug. 99	Fox	High definition	720p	24 hours	NA
KPRC (NBC)	Houston	Nov. 99	Post Newsweek	High definition	1080i	24 hours	NA
KTRK (ABC)	Houston	Jan. 00	ABC	High definition	720p	24 hours	NA
KOMO (ABC)	Seattle	Oct. 98	Fisher	High definition	720p	24 hours	NA
KCTS (PBS)	Seattle	Sept. 98	Public	High definition	1080i	24 hours	NA
KING (NBC)	Seattle	Oct. 98	Belo	High definition	1080i	24 hours	NA
KIRO (CBS)	Seattle	Nov. 99	Cox	High definition	1080i	24 hours	NA
WKYC (NBC)	Cleveland	June, 99	Gannett	High definition	1080i	24 hours	NA
WEWS (ABC)	Cleveland	Nov. 99	Scripps-Howard	High definition	720p	24 hours	NA
WJW (Fox)	Cleveland	Oct. 99	Fox	High definition	720p	varied hours	NA
WOIO (CBS)	Cleveland	Nov. 99	Raycom	High definition	1080i	24 hours	NA
WFLA (NBC)	Tampa./St. Pete	Nov. 99	NA	High definition	1080i	24 hours	NA
WTVT (Fox)	Tampa./St. Pete	March 2000	Fox	Standard definition	480p	24 hours	NA
WFTS (ABC)	Tampa./St. Pete	April, 2000	Gannett	NA	NA	NA	NA
WBAL (NBC)	Baltimore	Dec. 99	Hearst Argyle	NA	NA	NA	NA

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DATA COMING TO DTV

service to content providers looking to distribute data. The company plans to create a national infrastructure for delivering multimedia content to stations, using a combination of fiber optic lines and satellite capacity. It will build a network operations center and origination facility in Los Angeles and provide member stations with equipment that can receive Internet Protocol (IP) data, store it, and then insert it in local ATSC broadcast streams. Revenues will come solely from charging for the transport service--iBlast plans to let content providers collect their own advertising and/or subscription fees.

"We'll have about 60% of the country with an iBlast digital footprint by the first of next year," says iBlast President Ken Solomon, a former television syndication executive. Solomon says that the company is already talking to companies who could be "anchor tenants" for the service.

"It's wonderful because the demand is so high," he says.

Another consortium of stations pursuing a spectrum leasing agenda is the Broadcasters' Digital Cooperative, which was formed in late March by 12 station groups: Granite Broadcasting, Benedek Broadcasting, Capitol Broadcasting, Citadel Communications, Clear Channel Television, Cosmos Broadcasting (Cosmos has since defected to iBlast), Morgan Murphy Stations, Gray Communications, Nexstar Broadcasting, Pappas Telecasting, Paxson Communications, and Sunbelt Communications. Since then, several small groups

have come on board including Jefferson-Pilot Communications, Lamco Communications and Pegasus Broadcast Television. Each group pledged 4 Mb/s (daily average) out of each station's 19.4 Mb/s DTV bandwidth to the Cooperative. The group represents 196 stations in 130 markets reaching 85% of U.S. households.

The Cooperative's corporate structure is much less defined than iBlast, which signed long-term exclusive deals with its member groups. Its aim is simply to aggregate spectrum into a national footprint and then shop it to third parties who are willing to develop applications for the spectrum and pay to access it. Each company will receive an equity stake commensurate with its stations' coverage and each member of the Cooperative will help select executive staff and financial advisors to run the venture. The Cooperative has just reached an agreement with investment banking firm Salomon Smith Barney to represent the datacasting consortium in deals with wireless data companies.

When looking for potential customers to lease it spectrum, the Cooperative's first target is likely to be companies who have expressed an interest in the Ch. 60-69 UHF spectrum, which will be auctioned by the FCC to the highest bidders in September. According to the FCC's DTV rollout plan, the earliest that spectrum will be available is 2006. That's when the analog television spectrum is due to be returned to the government, provided that DTV has reached 85% penetration.

But the FCC is encouraging incumbent broadcasters and potential

bidders to find ways in which the spectrum could be vacated earlier. Since such agreements would obviously make the spectrum much more valuable, the FCC decided to postpone the 60-69 auction from its original June slot until September. In late June, the FCC also clarified some of the auction rules. One major change was to guarantee cable carriage for broadcasters on 60-69 who go digital quickly and give their analog spectrum back ahead of schedule. (The FCC has stated that the carriage rights will initially apply only to a station's primary digital signal and not to multicast digital signals.)

Another firm that wants to help broadcasters broker their digital spectrum is SpectraRep, a new company formed by longtime broadcast consulting firm BIA Financial Network. SpectraRep, based in Chantilly, Va., is asking stations to contribute 3 Mb/s of their DTV spectrum to the venture. SpectraRep will then pitch the spectrum to potential clients such as Internet content providers. In return, broadcasters will get a share in revenues and an equity stake in the venture.

"Stations will get compensated by how many people they cover, and how much data they distribute," says Tom Buono, President and CEO of BIA Financial Network. The company has already signed a deal with satellite capacity provider The Spaceconnection, which gives it a method to distribute data nationwide to local partner stations. ■

Glen Dickson is associate editor of Broadcasting & Cable, a publication of Cahners Business Information.



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Stations Broadcasting Digital Signal							
Station	City	Date On	Owner	Signal Type	Format	Time of Day	Cable Activity
WMAR (ABC)	Baltimore	Dec. 99	Scripps-Howard	High definition	720p	24 hours	NA
WBFF (Fox)	Baltimore	April, 2000	Sinclair	Standard definition	480p	24 hours	NA
WMPT (PBS)	Annapolis	April, 2000	Public	High definition	1080i	24hours	NA
KTCA (PBS)	Minn./St. Paul	Sept. 99	Public	High & Standard	1080i	24 varies	NA
KSTP (ABC)	Minn./St. Paul	Nov. 99	Hubbard	High definition	720p	24 hours	NA
WPLG (ABC)	Miami	Nov. 99	Washington Post	Test signal only	NA	NA	NA
WSVN (Fox)	Miami	Nov. 99	Independent	NA	NA	NA	NA
KPHO (CBS)	Phoenix	Nov. 99	Meredith	High & Standard	1080i, 480i	24 hours	NA
KSZA (Fox)	Phoenix	Nov. 99	NA	NA	NA	NA	NA
KNXV (ABC)	Phoenix	Jan. 00	Scripps-Howard	High definition	720p	24 hours	NA
KUTP (UPN)	Phoenix	March, 2000	United Television	Standard definition	480p	24 hours	NA
KATU (ABC)	Portland	Oct. 98	Fisher	High definition	720p	24 hours	NA
KOPB (PBS)	Portland	Nov. 98	Public	High definition	1080i	24 hours	NA
KPDQ (Fox)	Portland	Oct. 99	Meredith	Standard definition	480i	24 hours	NA
KOIN (CBS)	Portland	Dec. 99	Lee Enterprises	High definition	1080i	24 hours	NA
KGW (NBC)	Portland	Dec. 99	Belo	High definition	108i	24 hours	NA
KMGH (ABC)	Denver	NA	McGraw Hill	Test signal			NA
KDVR (Fox)	Denver	Oct. 99	Fox	High definition	720p	24 hours	NA
KRMA (PBS)	Denver	Dec. 99	Public	High definition	1080i	24 hours	NA
WTAE (ABC)	Pittsburgh	Jan. 99	Hearst Argyle	High definition	720p	24 hours	NA
WPXI (NBC)	Pittsburgh	Dec. 99	Cox	High definition	1080i	24 hours	NA
KCRA (NBC)	Sacramento	May, 99	Hearst Argyle	High definition	1080i	24 hours	NA
KTXL (Fox)	Sacramento	Nov. 99	Tribune	Standard definition	480p	24 hours	NA
KXTV (ABC)	Sacramento	Nov. 99	Gannett	High definition	720p	24 hours	NA
KMOV (CBS)	St. Louis	Nov. 99	Belo	High definition	1080i	24 hours	NA
KTVI (FOX)	St. Louis	Oct. 99	Fox	Standard definition	480p	24 hours	NA
KSDK (NBC)	St. Louis	Nov. 99	Gannett	High definition	1080i	24 hours	NA
WOFL (Fox)	Orlando	Jan. 2000	Meredith	Standard definition	480i	24 hours	NA
WTHR (NBC)	Indianapolis	Sept. 98	Dispatch	High definition	1080i	24 hours	NA
WISH (CBS)	Indianapolis	Dec. 98	LIN	High & Standard	1080i/480i	24 hours	NA
WXIN (Fox)	Indianapolis	Nov. 99	Tribune	Standard definition	480p	24 hours	NA
WRTV (ABC)	Indianapolis	Nov. 99	McGraw Hill	High definition	720p	24 hours	NA
KGTV (ABC)	San Diego	Nov. 99	McGraw Hill	High definition	720p	24 hours	NA
KNSD (NBC)	San Diego	Dec. 99	NBC	High definition	1080i	24 hours	NA
KFMB (CBS)	San Diego	May, 2000	Midwest Tel.	NA	NA	NA	NA
WBTV (CBS)	Charlotte	Nov. 98	Jefferson-Pilot	High definition	1080i	24 hours	NA
WSOC (ABC)	Charlotte	Oct. 99	Cox	High definition	720p	24 hours	NA
WCNC (NBC)	Charlotte	Nov. 99	Belo	High definition	1080i	24 hours	NA
WCCB (Fox)	Charlotte	Nov. 99	Bayhackle Comm.	NA	NA	NA	NA
WLWT (NBC)	Cincinnati	Nov. 98	Gannett	High definition	1080i	24 hours	NA
WCPO (ABC)	Cincinnati	Nov. 98	Scripps Howard	High definition	720p	10-6 (Mon.-Fri.)	NA
WKRC (CBS)	Cincinnati	Nov. 98	Jacor	High definition	1080i	Varies	NA
KSL (NBC)	Salt Lake City	Nov. 99	Bonneville Int.	High & Standard	480, 1080i	24 hours	NA
KTVX (ABC)	Salt Lake City	Nov. 99	United Television	NA	NA	NA	NA
WRAL (CBS)	Raleigh	May, 98	Capitol	High definition	1080i	24 hours	NA
WTVB (ABC)	Raleigh	Nov. 99	ABC	High definition	720p	24 hours	NA
WRAZ (Fox)	Raleigh	May, 2000	Capitol Brdcasting	High definition	1080i	24 hours	NA
WBNS (CBS)	Columbus	Nov. 98	Dispatch	NA	NA	NA	NA
WKOW (ABC)	Madison	Nov. 98	Shockley	High definition	720p	18 hours	NA
KITV (ABC)	Honolulu	Jan. 98	Hearst-Argyle	High definition	720p	24 hours	NA
WITF (PBS)	Harrisburg	Nov. 98	Public	High definition	1080i	24 hours	NA
KCPT (PBS)	Kansas City	Nov. 98	Public	High definition	1080i	24 hours	NA
WMVT (PBS)	Milwaukee	Oct. 98	Public	High & Standard		Varies	NA
WMPN (PBS)	Jackson	Nov. 98	Public	High definition	1080i	24 hours	NA
WTHH (ABC)	New Haven	Dec. 98	LIN	High definition	720p	24 hours	NA
WNOU (NBC)	South Bend	Dec. 98	Michiana Telecasting Corp.	High definition	1080i	Varies	NA
KXLY (ABC)	Spokane, WA	Sep. 99	Morgan Murphy	High definition	720p	24 hours	NA
WMFD (Ind.)	Mansfield, OH	Feb. 99	Independent	Standard definition	480i	24 hours	NA
WKPC (PBS)	Louisville, KY	Aug. 99	Public	High definition	1080i	18 hours	NA
WOOD (NBC)	Grand Rapids	Aug. 99	Lin Television	High definition	1080i	24 hours	NA
WCCB (PBS)	Lewiston, ME	Nov. 99	Public	High definition	1080i	24 hours	NA
KNTV (ABC)	Monterey, CA	Dec. 99	Granite Brdcasting	High definition	720p	24 hours	NA
WRDW (CBS)	Augusta, GA	March 2000	Gray Comm.	High definition	720p	24 hours	NA
WRLK (PBS)	Columbia, SC	March, 2000	Public	High & Standard	480/1080i	24 hours	Digital signal carried on Time-Warner system
KLAS (CBS)	Las Vegas, NV	April, 2000	Landmark Comm.	High definition	1080i	24 hours	NA
WGBY (PBS)	Springfield, MA	March, 2000	Public	High & standard	480p/1080i	12 hours	NA
KFOR (NBC)	Oklahoma City	May, 2000	New York Times Co.	NA	NA	NA	NA
WCYB (NBC)	Bristol, TN	May, 2000	Lamcro Comm.	High definition	1080i	24hours	NA